

the majority of patients respond positively to three or four sessions of palliative radiation therapy generally spaced a week apart, the likelihood of fracture of the affected bone increases after the therapy. The rehabilitation programs of geriatric cancer patients are designed to be gentle, low-intensity exercises and manipulation to minimize the stress to the patient.

Summary

The optimal rehabilitation of geriatric patients involves a thorough initial patient assessment, focusing on the identification of underlying musculoskeletal and metabolic disorders, weight management, OA management (if necessary), environmental modifications that promote low-stress activities, and patient assessment by the veterinarian at regular time intervals to monitor patient progress.

REFERENCES

- Hoskins JD: Geriatrics, *Vet Clin North Am Small Anim Pract* 27:1273-1284, 1997.
- Goldston RT: Geriatrics and gerontology, *Vet Clin North Am Small Anim Pract* 19:1-202, 1989.
- Kealy RD et al: Effects of diet restriction on life span and age-related changes in dogs, *J Am Vet Med Assoc* 220:1315-1320, 2002.
- Goldston RT, Hoskins JD: *Geriatrics and gerontology of the dog and cat*, Philadelphia, 1995, Saunders.
- Tonner PH, Kampen J, Scholz J: Pathophysiological changes in the elderly, *Best Pract Res Clin Anaesthesiol* 17:163-177, 2003.
- Fleming JM, Creevy KE, Promislow DEL: Mortality in North American dogs from 1984 to 2004. An investigation into age-, size-, and breed-related causes of death, *J Vet Intern Med* 25:187-198, 2011.
- Laine L, Curtis SP, Cryer B et al: Risk factors for NSAID-associated upper GI clinical events in a long-term prospective study of 34701 arthritis patients, *Aliment Pharmacol Ther* 32:1240-1248, 2010.
- Joubert KE: The effects of firocoxib (Previcox) in geriatric dogs over a period of 90 days, *J South Afr Vet Assoc* 80:179-184, 2009.
- Bromiley MW, Ulbrich M: *Physiotherapy in veterinary medicine*, ed 1, Oxford, UK, 1995, Blackwell Scientific.
- Downer AH, Spear VL: Physical therapy in the management of long bone fractures in small animals, *Vet Clin North Am* 5:157-164, 1975.
- Millis DL, Levine D: The role of exercise and physical modalities in the treatment of osteoarthritis, *Vet Clin North Am Small Anim Pract* 27:913-930, 1997.
- Minor MA, Gewett JE, Webel RR: Exercise and disease related measures in patients with rheumatoid arthritis and osteoarthritis, *Arthritis Rheum* 32:1396, 1988.
- Nicholas JJ: Physical modalities in rheumatological rehabilitation, *Arch Phys Med Rehabil* 75:994, 1994.
- Baker K, Grainger A, Niu J et al: Relation of synovitis to knee pain using contrast-enhanced MRIs, *Ann Rheum Dis* 69:1779-1783, 2010.
- Millis DL, Loonam JE, Evans M: *Comparison of carprofen and etodolac for the management of stifle osteoarthritis: a pilot study*. Proceedings of the Veterinary Orthopedic Society Annual Meeting, Steamboat Springs, CO, 2003.
- Roush JK, Cross AR, Renberg WC et al: Evaluation of the effects of dietary supplementation with fish oil omega-3 fatty acids on weight bearing in dogs with osteoarthritis, *J Am Vet Med Assoc* 236:67-73, 2010.
- Roush JK, Dodd CE, Fritsch DA et al: Multicenter veterinary practice assessment of the effects of omega-3 fatty acids on osteoarthritis in dogs, *J Am Vet Med Assoc* 236:59-66, 2010.
- Phitak T, Pothacharoen P, Kongtawelert P: Comparison of glucose derivatives effects on cartilage degradation, *BMC Musculoskelet Disord* 11:162, 2010.
- Derfoul A, Miyoshi AD, Freeman DE et al: Glucosamine promotes chondrogenic phenotype in both chondrocytes and mesenchymal stem cells and inhibits MMP-13 expression and matrix degradation, *Osteoarthritis Cartilage* 15:646-655, 2007.
- Millis DL et al: Effect of polysulfated glycosaminoglycan on articular cartilage in a Pond-Nuki model of osteoarthritis, *Vet Surg* 27:293, 1998.
- Felson DT, Zhang Y, Anthony JM: Weight loss reduces the risk for symptomatic knee osteoarthritis in women, *Ann Intern Med* 116:535, 1992.
- Marshall WG, Bockstahler BA, Hulse DA et al: A review of osteoarthritis and obesity: current understanding of the relationship and benefit of obesity treatment and prevention in the dog, *Vet Comp Orthop Traumatol* 22:339-345, 2009.
- Schrager M: Slimming down reduces arthritis risk, *Phys Sportsmed* 23:22, 1995.
- McGoey BV, Deitel M, Saplys RJ: Effect of weight loss on musculoskeletal pain in the morbidly obese, *J Bone Joint Surg* 72B:323, 1990.
- Impellizzeri JA, Tetrack MA, Muir P: Effect of weight reduction on clinical signs of lameness in dogs with hip osteoarthritis, *J Am Vet Med Assoc* 216:1089-1091, 2000.
- Burkholder W, Hulse DA: *Signs of osteoarthritis can be improved with weight loss and exercise*, St. Louis, 2002, Nestle Purina Pet Care.
- LaFlamme DP: Development and validation of a body condition score system for dogs, *Canine Practice* 22:10-15, 1997.
- Felson DT: The epidemiology of knee osteoarthritis: results from the Framingham osteoarthritis study, *Semin Arthritis Rheum* 20:42, 1990.
- Jurvelin J et al: Effect of physical exercise on indentation stiffness of articular cartilage in the canine knee, *Int J Sports Med* 7:106-110, 1986.
- Kiviranta I et al: Moderate running exercise augments glycosaminoglycans and thickness of articular cartilage in the knee joint of young beagle dogs, *J Orthop Res* 6:188-195, 1998.
- Newton PM et al: The effect of lifelong exercise on canine articular cartilage, *Am J Sports Med* 25:282-287, 1997.
- Arokoski J et al: Softening of the lateral condyle articular cartilage in the canine knee joint after long distance (up to 40 km/day) running training lasting one year, *Int J Sports Med* 15:254-260, 1994.
- Hallett MB, Andrich JT: Effects of exercise on articular cartilage, *Sports Med Arthrosc Rev* 2:29-37, 1994.

34. McKeag DB: The relationship of osteoarthritis and exercise, *Clin Sports Med* 11:471, 1992.
35. Kovar PA, Allegrante JP, Mackenzie CR: Supervised fitness walking in patients with osteoarthritis of the knee: a randomized, controlled trial, *Ann Intern Med* 116:529, 1992.
36. Lewis C: Arthritis and exercise. In Biegel L, ed: *Physical fitness and the older person: a guide to exercise for health care professionals*, ed 1, Rockville, MD, 1984, Aspen Systems.
37. Bender T, Nagy G, Barna I et al: The effect of physical therapy on beta-endorphin levels, *Eur J Appl Physiol* 100:371-382, 2007.
38. Halbertsma JP, van-Bolhuis AI, Goeken LN: Sport stretching: effect on passive muscle stiffness of short hamstrings, *Arch Phys Med Rehabil* 77:688-692, 1996.
39. Magnusson SP: Passive properties of human skeletal muscle during stretch maneuvers: a review, *Scand J Med Sci Sports* 8:65-77, 1998.
40. Johnston KD, Levine D, Price MN et al: *The effect of TENS on osteoarthritic pain in the stifle of dogs*. Proceedings of the 2nd International Symposium on Physical Therapy Rehabilitation Veterinary Medicine, Knoxville, TN, 2002.
41. Baird CL: First-line treatment for osteoarthritis. Part 2: Non-pharmacologic interventions and evaluation, *Orthop Nurs* 20:13-18, 2001.



36 Physical Rehabilitation for Wound Care

June Hanks and Sallye Gregg

Patients undergoing rehabilitation frequently have wounds, including surgical wounds and wounds resulting from nonsurgical trauma. Therefore, an understanding of wound healing and the care of wounds is necessary to speed the rehabilitation process and identify abnormal wound healing. Normal restoration of injured tissue follows a series of predictable overlapping events often grouped as the inflammatory, proliferation, and maturation/remodeling phases of wound healing (Figure 36-1). Biochemical substances modulate cellular activities occurring during the phases of tissue repair. Acute wounds tend to follow a predictable progression through the wound healing phases, whereas chronic wounds may have a prolonged period of inflammation, proliferation, or excessive wound contraction.

This chapter reviews the wound healing phases and discusses factors that may delay the tissue restoration process. Modalities, dressings, and alternative treatments to speed the healing process are discussed.

Wound Healing Physiology

Inflammatory Phase

The first response to injury is inflammation, a process that includes vascular and cellular events to control blood loss, combat bacterial invasion, and set the stage for further healing by signaling the cells necessary for repair and regeneration to come to the site of injury. For a few seconds to several minutes following injury, blood vessels in the area reflexively constrict to reduce blood loss. Platelets aggregate at the injury site and are activated by contact between damaged endothelial cells lining the blood vessel walls and exposed collagen. The activated platelets adhere to one another and the injured endothelium and interact with fibrin to help form a plug to wall off the injured area. The activated platelets also release chemical mediators such as cytokines (signaling proteins), growth factors (hormone-like substances that control cell metabolism), and chemotactic agents (substances that attract cells important for repair to come to the site of injury).^{1,2}

Following vasoconstriction blood vessels dilate, allowing more fluid, inflammatory cells, and growth factors into the interstitial space. The walls of injured blood vessels allow fluid to leak into the interstitial space, causing

localized edema that serves to increase pressure in the interstitial space and decrease pressure in the blood vessels. As the wound area accumulates more proteins, white blood cells, and cell fragments, the fluid in the interstitial space becomes more viscous and changes from a transudate to an exudate. Mast cells in the area release inflammatory mediators and secrete enzymes. Vasodilation is induced by histamine released by mast cells and prostaglandins released by injured cells.³ Swelling and chemical release in the area continue to increase, causing pain. The entire series of events leads to the hallmark signs of inflammation: edema, heat, redness, pain, and decreased function.

Cellular events that accompany the vascular events include margination of neutrophils to the sides of the blood vessel wall and eventual migration through the wall. Guidance to the injured area occurs as neutrophils follow the chemical gradient formed by bacteria and dead or dying cells through a process called *chemotaxis*. Circulating monocytes migrate into the tissue spaces and are then called *macrophages*. Neutrophils and macrophages release enzymes, engulf bacteria, produce tissue proteases that further degrade debris, and release chemotactic agents and chemical mediators that attract more inflammatory cells to the area. Although less effective than neutrophils in the phagocytic process, macrophages produce numerous growth factors and are critical in directing the tissue repair process.^{4,5}

Proliferative Phase

As the cells necessary for tissue repair respond to signals to enter the injured area, the proliferative phase of wound repair begins. Endothelial cells adjacent to the site of injury respond to ischemia, growth factors, and chemical mediators and begin to bud into the injured area through the process of angiogenesis. As buds connect through capillary loops, nutrients are carried to the area and waste products are removed. The blood vessel buds appear as tiny red dots in the tissue space. Previously damaged lymphatic vessels begin to function again and work with the blood vessels to reduce local tissue edema. Proteases produced by various cells work alongside phagocytic cells to degrade wound debris.

Fibroblasts from the local tissue and general circulation begin to fill the spaces between the capillary loops. As the fibroblasts proliferate among the capillary loops, an

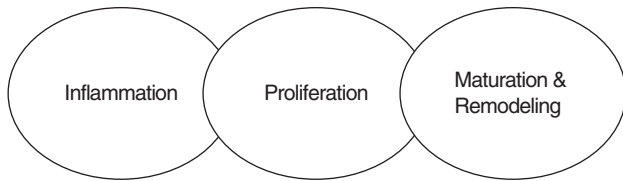


Figure 36-1 Overlapping phases of wound healing.



Figure 36-2 Granulation tissue covering a large wound.

extracellular matrix composed of water, proteoglycans, and other cells is formed. The matrix fills the space between collagen and elastin fibers in the wounded area, allowing more fibroblasts to migrate across the wound bed. The newly formed vascularized connective tissue is called granulation tissue (Figure 36-2). Chemotactic agents and oxygen levels guide fibroblasts to migrate and proliferate, thus filling in the wound bed. Integrins, surface cell receptors, serve as chemotactic agents and adhesion molecules to assist the migration of fibroblasts within the extracellular matrix.¹ As oxygen levels increase in the wounded area, less proliferative activity is stimulated. Some of the fibroblasts transform into myofibroblasts which have contractile properties and begin to pull the wound edges inward. The wound contraction decreases the size of the defect. In larger and deeper defects, more granulation tissue is needed to fill the void calling for a greater degree of wound contraction. The shape of the wound affects the speed of contraction. Linear wounds contract faster than rectangular wounds. The slowest to contract are circular wounds.

Epithelial cells (primarily keratinocytes) migrate across the granulation tissue bed from the wound margins and epidermal appendages to cover the wound defect. Growth factors stimulate epithelial cell proliferation while oxygen levels and chemotaxis guide migration across the wound bed. Ease of migration is facilitated in a moist wound environment.⁶ After the wound has been completely resurfaced with epithelial cells, the proliferative phase is considered complete.

Maturation/Remodeling Phase

Although the wound bed is resurfaced through the cellular and vascular events in the inflammatory and proliferative

phases, the wound is not yet completely healed. The newly formed tissue must be strengthened and reorganized through collagen synthesis and lysis in the newly formed scar tissue. Through chemical events, internal stress, and external tension, the collagen fibers transform from immature type III collagen to mature type I collagen.⁷ Interventions of range of motion exercise and scar tissue mobilization aid the process of scar tissue remodeling eventually achieving a balance between collagen formation and degradation to produce a strong healthy scar. Most of the remodeling of the scar occurs in the first 6 to 12 months after wound closure, but can continue for up to 2 years. Even when fully remodeled, the scar tissue attains only 75% to 80% of the strength and elasticity of the original tissue.⁸

Types of Wound Closures

Wounds that heal following approximation of the wound edges with little or no granulation tissue formation (as in a surgical wound) are said to heal by primary closure or primary intention. An uncontaminated, clean wound bed with wound margins that can be brought together with little tension on the surrounding tissues is required for healing by primary closure. Healing by secondary closure, or secondary intention, occurs when the wound edges cannot be brought together. The process of granulation tissue formation, wound contraction, and epithelialization is required to fill the wound defect. Healing by secondary closure is more complicated, results in more scar tissue formation, and requires more time and energy to achieve closure than does healing by primary intention. In some cases, a traumatic or surgical wound cannot be immediately closed because of contamination or risk of infection. The wound may be cleaned over a period of a few days. When the wound is free of debris, it may be cleaned, surgically debrided and closed. Such a wound is said to heal through delayed primary closure.

Abnormal Wound Healing

A variety of factors may result in failure of an acute wound to move through the inflammatory, proliferative, and maturation phases of healing in a predictable fashion. Wounds may become chronic because of a larger number of proteases and lack of protease inhibitors, leading to a degradation of the extracellular matrix and a larger wound bed. Chronic wounds have higher amounts of inflammatory immunomodulating agents and are less responsive to growth factors. Patients who are immunocompromised or who have arterial insufficiency may be unable to mount the inflammatory response necessary to clean the wound of debris and produce chemotactic cells that signal other cells necessary for the production of granulation tissue in the wounded area. Such wounds may be void of the classic signs of inflammation. In contrast, some wounds may persist in the inflammatory phase, exhibiting debris, rolled,

Table 36-1 Factors Affecting Wound Healing

Wound Characteristics	Local Factors	Systemic Factors	Improper Management
Time since onset	Circulation	Age	Allow wound to become too wet or too dry
Location	Sensation	Inadequate nutrition	Failure to fill wound cavity with dressing
Dimensions	Mechanical stress	Co-morbidities	Failure to thoroughly cleanse wound
Temperature		Medication	Overuse of cytotoxic agents
Hydration			
Necrotic tissue			
Infection			

Myers BA: *Wound management: principles and practice*, ed 2. Englewood Cliffs, NJ, 2008, Pearson Prentice Hall.

thickened wound edges, and a lack of adherence of the wound edge to the wound bed. Common causes of chronic inflammation include the presence of a foreign body in the wound bed, repetitive trauma (e.g., excessive licking of the wound bed), and the overuse of agents that are cytotoxic to the cells involved in the healing process. Currently there are insufficient data to assess safety, efficacy, and recommendations for use of some topical agents because of inconsistency or lack of verification of equivalent comparisons between in vitro and in vivo studies.⁹ Agents in question include povidone-iodine, hydrogen peroxide, sodium hypochlorite bleach, and chlorhexidine.¹⁰⁻¹²

Some wounds may granulate too little (hypogranulate) because of poor support for a granular matrix. A “pothole” wound is formed as epithelial cells pile up at the wound edge and fail to migrate across the wound bed. The wound edges of such pothole wounds should be gently wiped with gauze and the wound cavity should be lightly filled with a dressing to promote the formation of a granular matrix and prevent premature migration of epithelial cells. In contrast, hypergranulation tissue may be formed when a wound defect has been filled, yet the granulation process continues, resulting in granulation tissue that extends well above the wound surface. Hypergranulation tissue, or *proud flesh*, can be resolved by the application of pressure to the tissue through certain dressing types, such as hydrocolloid dressing, the use of silver nitrate to cauterize the tissue and level the wound bed, hypertonic saline dressings, or surgical excision of excess tissue.

Factors Affecting Wound Healing

Deterrents to expedient wound healing include wound characteristics, local factors, and systemic factors (Table 36-1). Acute wounds and surgically induced wounds tend to heal more quickly than severe traumatic wounds or wounds of insidious onset. Wounds located over bony prominences, in areas with few epidermal appendages, across joints, and in areas of thick skin have slower healing times. Because more cellular activity and rebuilding of tissue are required, larger and full-thickness wounds require more healing time than smaller or more

superficial wounds. The close proximity of wound edges results in shorter healing times for linear wounds than for rectangular or circularly shaped wounds. Maintenance of appropriate wound moisture within the wound bed encourages granulation tissue formation and epithelialization. Thus, wound beds and periwound areas that are too dry (desiccated) or too wet (macrated) have prolonged healing times. Devitalized tissue within the wound bed and dry exudate or scabs can slow the migration of epithelial cells across the wound bed, prolong the inflammatory response, and create an environment conducive to infection.^{13,14}

Important to consider is the fact that all wounds are contaminated. All body surfaces, including skin, and the digestive tract have a normal microflora of bacteria and fungi that include *Staphylococcus*, *Micrococcus*, *Peptococcus*, *Streptococcus*, *Acinetobacter*, and yeast. The resident microflora and pH of the skin create a slightly acidic environment, serving a protective function against disease-causing microorganisms.¹⁵ In an open wound the microflora adhere to the surface and replicate, resulting in colonies that do not necessarily adversely affect the host as the normal immune response is activated. However, if the bacteria or fungi continue to replicate, a point of critical colonization is reached in which the microorganisms become a bioburden, adversely affecting the host. When microorganisms multiply to the point of invading viable body tissues, the wound is considered infected. Because of the presence of individual factors in each host, there is no specific threshold for infection, although in general, wounds with a bacterial quantity of greater than 10^5 microbes per gram of tissue are considered infected.¹⁶ However, the type of microorganism must also be considered because some are more virulent and produce exotoxins that may inactivate or modify host responses, leading to host cell dysfunction or death. Bacterial endotoxins, molecules within the walls of gram-negative bacteria, may induce a strong inflammatory response leading to fever, bleeding, and clotting. In addition interactions among bacteria can lead to a microorganism community called a biofilm that attaches to the wound surface and is encased in a self-developed glycocalyx

(which protects the bacteria). The microorganisms encased in the biofilm exhibit increased virulence and resistance, making them less responsive to biological and pharmaceutical intervention.¹⁷ Immunocompromise, malnutrition, and impaired local circulation of the host can also affect resistance to invading organisms.

In general, infection should be suspected when the signs of inflammation are disproportionate to the size and extent of the wound (e.g., drainage is copious, has uncharacteristic color, wound is very painful, periwound area is extremely red or hot, etc.). Wound cultures such as tissue biopsies, swab cultures, or fluid aspirations may be used to identify an offending microorganism. Although tissue biopsy is the gold standard for culturing, properly performed swab cultures are a reliable method to identify and quantify bacteria in a wound.¹⁸ To properly perform a swab culture, the clinician must first debride and thoroughly rinse the wound bed. The swab should then be moved slowly over viable tissue in the wound bed in a crisscross pattern back and forth, moving from top to bottom while rolling the swab and applying enough pressure to express some of the wound bed fluid. Wounds with a small surface area of little granulation tissue should be cultured by swabbing only the small area for 5 seconds while rotating the applicator.¹⁹ Tissue biopsies are preferred for evaluating pressure/decubital ulcers, burns, or wounds with a suspected neoplastic process.

Because endogenous or exogenous microbes can lead to infection, meticulous adherence to infection control procedures must be exhibited. Procedures include hand washing before and after glove use, the use of gloves, which should be changed between procedures on the same patient and promptly after touching contaminated materials, and isolation of patients with antibiotic-resistant infections.

Emergency Care

In caring for a patient with a wound, the safety of the patient and staff is of upmost concern. When in pain, pets can become suddenly aggressive. Staff should have muzzles and blankets readily available for use as necessary. A muzzle can be used during the administration of parenteral pain medications and placement of an intravenous (IV) line if patient respirations are normal. For patients having problems breathing, a thick blanket can be placed over the animal to control the head during the evaluation and administration of pain medications. It is necessary to exercise universal precautions including the use of gloves to protect the provider from human blood that may be present on the coat of the patient and to protect the wounded area of the patient from bacteria that may be spread by health care workers.

Medications to relieve pain should not be withheld from a patient, because analgesic agents do not prevent a full patient evaluation from being completed. Pain medications do not mask the physiologic indicators, because the

heart rate will still be elevated in response to hypovolemia, hypoxia, hypotension, or hypercarbia, even when high doses of opioid medications have been delivered to the patient. In the emergency patient opioids are unlikely to cause respiratory depression or the deterioration of hemodynamic parameters.²⁰ The use of nonsteroidal antiinflammatory drugs (NSAIDs) and corticosteroids are contraindicated for use in a patient in shock until renal perfusion, blood pressure, and dehydration issues have been addressed. In addition, endogenous endorphins released in a stressful situation, such as an emergency room, can mask the seriousness of the patient's condition; therefore chemical sedation for fear and aggression is a humane way to examine the emergency patient. An oxygen mask or oxygen induction chamber allows the patient to relax while medications are administered.²¹

Severity of the injury determines the proper medications for pain control. Severe to excruciating pain present after trauma with fractures and muscle damage require high-dose opioids and additional pain medications for adequate relief. Fractures should be immobilized as soon as possible to reduce pain. Muscle hemorrhage, hematomas, or bruising may cause significant discomfort. These injuries may benefit from local application of ice during the first 72 hours or until the acute inflammatory phase has ended, and heat application thereafter. For injuries to the perineum, pelvis, and rear limbs, epidural anesthesia may be necessary if opioids are inadequate to control pain. After the patient has been stabilized, adjunctive medications such as NSAIDs, medications for nerve pain, and low-dose sedatives may be administered.¹

Nursing care is extremely important in controlling pain and anxiety. The patient should be kept clean, warm, and dry on thick padded bedding. The extra attention and care shown to the patient may decrease the anxiety level and lessen the length of time the patient must be hospitalized.

Initial Assessment

Although external bleeding wounds often capture the attention of the owners, the health care worker must be aware that more life-threatening injuries may be present. The initial stabilization includes an examination of the airway with administration of oxygen, as appropriate. In the case of severe head wounds, facial trauma or tracheal damage, rapid induction may be necessary to accomplish intubation and obtain an open airway (Figures 36-3 and 36-4). Puncture wounds or trauma to the chest wall should increase suspicion of a pneumothorax, for which a thoracentesis or chest tube may be necessary. All vital signs should be obtained to evaluate the need for IV fluids and/or blood products to treat hypovolemic shock.

Bleeding from a wound may be controlled following or simultaneously with establishing an adequate airway and stabilizing hemodynamic parameters. Capillary bleeding and venous hemorrhage normally can be controlled with gauze sponges, direct pressure, and a bandage. With



Figure 36-3 Puncture wounds over the neck of a dog that sustained bite wounds.

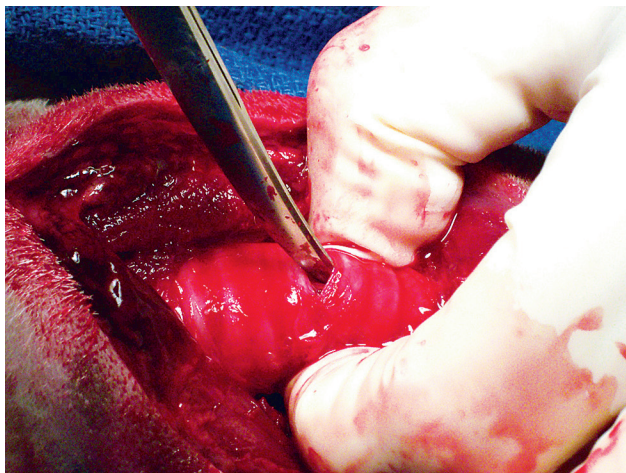


Figure 36-4 Exploration of the wound in Figure 36-3 resulted in finding a laceration of the trachea.

a gloved hand, a sterile gel should be placed in the wound and the wound should be covered with a layer of gauze sponges. Compression over the gauze sponges may be maintained with a layer of elastic bandage or tape until the wound can be definitively addressed. More aggressive therapy is needed for arterial bleeding. For appendicular wounds, a 5- to 10-cm wide blood pressure cuff can be applied proximal to the wound. The cuff is inflated to 20 to 30 mm Hg higher than the systemic arterial pressure to control the bleeding while sedation is given. The inflated cuff can serve as a temporary tourniquet for 30 to 45 minutes while the patient is sedated and prepared for surgical exposure to the artery. A sterile hemostat is applied to the bleeding vessel and ligation with a small diameter antimicrobial suture is performed. Even though the blood pressure cuff can be left in place for an extended period without permanent damage to the limb, the pressure should be released every 5 to 10 minutes to allow blood flow to and from the region. Commercial tourniquets made from narrow bands of elastic material or Penrose drains should not be used because they can place excessive pressure on neurovascular structures and cause permanent damage.

After life-threatening injuries have been addressed, a thorough examination of the wounded area should be

conducted. Traumatic wounds are classified according to type and contamination. Wound types include lacerations, avulsions, shearing injuries, punctures or bites, penetrating wounds, and pressure necrosis (Table 36-2). Wounds also may be classified based on degree of contamination (Table 36-3).

The initial lavage, debridement, and aftercare are to encourage healthy tissue that is free of infection and necrosis, and has a good blood supply. The golden period of wound care refers to the first 6 hours after the traumatic event occurred. It is during this time that the wound may be cleaned and closed without the development of infection. While stabilization is occurring, the wound should be kept as moist as possible by placing sterile water-soluble gel into the wound and covering with a saline-soaked sponge. Wounds that are kept moist heal up to 50% faster than wounds that are allowed to dry out. Wide clipping of the wound enables the clinician to examine the injury and often exposes wounds that were not apparent upon initial inspection. In the case of a bite wound, the opposite side of the affected limb or body part should be carefully examined for a potential matching injury from the rest of the teeth involved in the bite.

The wounded area must be cleansed of all debris and blood. In a grossly contaminated wound, tap water is practical to use for the first cleaning. Obvious debris can be washed away with warm water using a gentle spray attachment. A surgical scrub solution should then be applied to the intact skin and the area around the wound. Detergent-containing scrubs are cytotoxic and may delay healing, especially if used for extended periods of time. An exception is the topical microbicide chloroxylenol (brand name Techni-Care®), which, in a retrospective clinical trial, appeared to decrease infection rates and allow the wound to progress toward healing²² (Table 36-4).

Following the removal of gross contamination and cleansing of the surrounding skin, the wound must be lavaged with large amounts of fluid (at least 500 mL). The statement, “The secret to treating pollution is dilution” truly applies to initial wound management. Fluid can be sprayed on the wound using an 18-gauge catheter attached to a 35-mL syringe via a three-way stopcock. A warm 1-L bag of fluid is also attached, enabling the operator to rapidly fill the syringe and flush the wound. This mechanism provides a pressure of 7 to 9 pounds per square inch (PSI), which is sufficient for decreasing the bacterial population, but not so high as to drive the bacteria deeper into the tissues. Room temperature or cold lavage solutions are uncomfortable and possibly painful to the patient. The least toxic, most inert lavage solution is ideal. The topical application of a mixture of sugar or honey, sugar and povidone-iodine (PI), or honey and PI may be used to promote granulation tissue formation and reduce bacteria in the wound. Used alone or in combination with PI, sugar acts to inhibit bacterial growth and modulates keratinocytes and fibroblasts.^{23,24}

Table 36-2 Types of Traumatic Wounds

Laceration	The wound exhibits minimal necrosis or compression and has sharp, linear, or jagged edges. Normally this is a relatively clean injury that requires minimal debridement, but inspection of the deeper tissues is important.
Avulsion	The injury results from tensile forces pulling the tissue free of its natural attachments. Deep exploration with delayed closure is beneficial for this type of wound.
Shearing or degloving	The high friction of the pavement on the skin during a motor vehicle accident will often result in tearing of the skin and its attachments as well as damage to the deep vascular, nerve, and bony structures. These wounds have minimal to moderate contamination and benefit from delayed closure. Skin grafts and orthopedic repairs are common with this injury.
Punctures	The small skin wound is only the tip of the iceberg in this injury. The compressive forces and trauma to the underlying tissues from a bite injury involves significant disruption of the deep tissue planes as well as deep seeding of bacteria into the wound. Foreign bodies that range from hair to pieces of sticks and gravel are often found in these wounds. The wound should be opened and explored to clean, debride, and alleviate dead space. Delayed closure and antibiotics are beneficial in these patients.
Penetrating injuries from projectiles	These wounds are considered to be contaminated and require surgical exploration if the wound is over the abdominal cavity or the bullet is located within a joint.
Pressure necrosis	Devitalized necrotic tissue occurring from sustained pressure causing ischemia such as from a tightly wrapped bandage, lying for extended time periods on a bony prominence, or from crushing and vascular injury during a trauma.

Halling K: Wounds and open fractures. In Mathews KA, editor: *Veterinary emergency and critical care manual*, ed 2. New York, 2007, Lifelearn; Swaim SF, Henderson RA: *Small animal wound management*. Philadelphia, 1990, Lea & Febiger.

Table 36-3 Wound Classification/Contamination

Clean	Injury created by a grossly clean object and is less than 6 hours old
Contaminated	Created by an object with bacterial contamination and/or is greater than 6 hours old
Dirty/infected	Old traumatic wound with purulent material or necrotic tissue present

Dernell WS: Initial wound management. *Vet Clin North Am Small Anim Pract* 36(4):714-773, 2006; Garzotto CK: Wound management. In Silverstein DC, Hopper K, editors: *Small animal critical care medicine*. St. Louis, 2008, Saunders.

Principles of Wound Management

Basic principles of wound management include cleaning the wound, preventing infection, providing a moist wound environment, and attending to systemic needs. Provision of a moist, but not wet, wound environment leads to increased angiogenesis, decreased dehydration and cell death, enhanced autolytic debridement, increased epithelialization, and the provision of a bacterial barrier to decrease infection rates. The appropriate choice, technique

Box 36-1 Wound Management Techniques to Create and Maintain a Moist, Clean Wound Environment

Dirty wound?	→	Clean it
Cavity/hole?	→	Fill it
Wet wound?	→	Use more absorptive dressing/change more frequently
Dry wound?	→	Use less absorptive dressing
Infected wound?	→	Address underlying cause, clean wound, use antimicrobial dressing

of application, and frequency of dressing changes to maintain a moist wound bed facilitates granulation tissue formation and migration of epithelial cells across the wound bed (Box 36-1).

Wound Dressings

The ideal wound dressing provides an optimal moist wound healing environment, thermal insulation, gaseous exchange, pain relief, edema control, dead space elimination, protection from contaminants, and assistance with removal of dead tissue without disruption of the wound bed. The primary dressing or contact layer may be adherent or nonadherent, although in general nonadherent contact layers are recommended to prevent disruption of

Table 36-4 Properties of Surgical Scrubs

	Chlorhexidine	Povidone-Iodine	Alcohol	Techni-Care®
Length of cidal Capacity	>6 hr	3 hr	None	>6 hr
Ocular irritation	Yes	Yes	Yes	Minimal
Scrub time	8 min	3.5 min	11 min	0.5 min
Amount needed	10 mL	10 mL	N/A	2.6 mL
Tissue contraindication	Yes	Yes	Yes	None
Toxicity	Skin, eyes, ears, genitals	Skin	Skin, eyes	Nontoxic
Denatured by organic material	Minimal	Yes; inactivated by blood	No data	Minimal
Transdermal penetration	No	Yes	No	Yes
pH	5.84	4.0	6.8	7.2

Grubbs BC, Statz CL, et al.: Salvage therapy of open infected surgical wounds: a retrospective review using Techni-Care®. *Surg Infect* 1(2):109, 2000; Statz CL: Fighting wound infections. *Plastic Surg Prod* 11:34-40, 2005.

**Figure 36-5** Various dressings used over multiple wounds.

granulating tissue when dressings are removed. Permeability and occlusive properties vary among dressings. Semi-occlusive dressings are most commonly used to create a moist wound environment, allowing air to pass through the dressing and exudate to be absorbed into the dressing from the wound bed. The secondary dressing is placed over the primary dressing to increase protection, cushioning, absorption, or occlusion and may be used to hold the primary dressing in place²⁵ (Figure 36-5). Major dressing types include gauze, alginate, films, foam, hydrogels, and hydrocolloids (Table 36-5).

Other Wound Healing Therapies

Hyperbaric Oxygen Therapy

Hyperbaric oxygen therapy (HBOT), traditionally used to treat decompression illness (the bends), has been used for some time in wound care. Only minimal increases in

**Figure 36-6** Small animal hyperbaric oxygen chamber.

plasma oxygen levels occur with oxygen delivery by mask or other indirect routes, whereas large increases of dissolved oxygen in the plasma occur with oxygen delivery under increased atmospheric pressure. Compared with normobaric conditions, the increase in partial pressure of dissolved oxygen in plasma and the increase in delivery of oxygen to the tissues provides an environment conducive for healing of ischemic or otherwise hypoxic tissues. Angiogenesis is stimulated, immune system function is improved, and proliferation is enhanced.²⁶

Hyperbaric oxygen therapy chambers come in various sizes and can accommodate more than one small animal per treatment session (Figure 36-6). Each HBOT session is managed by trained personnel. Contraindications include deep vein thrombosis and congestive heart failure. Relative contraindications include severe arterial deficiency, chronic obstructive pulmonary disease, and pregnancy.

Table 36-5 Wound Dressings

Type of Dressing	Absorption Capacity	Benefits	Limitations	Common Uses
Gauze	Low (moderate if multiple layers are used)	Inexpensive	May adhere to wound bed; require more frequent dressing changes	Infected wounds; wound requiring packing; wounds requiring frequent dressing changes
Hydrogels	Low to moderate	Moisture retentive; encourage autolytic debridement	Cannot be used on highly exudative wounds; should not be used on infected wounds; generally require a secondary dressing	Soften eschar and provide moisture for dry wounds
Semipermeable foam	Moderate	Moisture retentive; may be left in place for several days; provides cushioning and insulation	Should not be used on infected wounds unless changed frequently; may roll in areas of friction	Minor burns; skin grafts; pressure ulcers
Hydrocolloid	Moderate	Impermeable to urine, stool, bacteria; provides thermal insulation	May traumatize periwound area on removal; may cause hypergranulation; should not be used on infected wounds	Pressure ulcers; burns
Alginate	High	Can be used on infected or uninfected wounds; are nonadherent and are biocompatible	Require secondary dressing; should not be used on dry wounds	Heavily exuding wounds
Silver-impregnated dressing	Varies, dependent on base dressing (e.g., hydrocolloid, alginate, foam, gel, etc.)	Broad spectrum antimicrobial properties	Distribution of silver within dressing and release of silver into wound greatly affects antimicrobial activity	To reduce bioburden in wound, reduce risk of infection

Electrical Stimulation

The effectiveness of electrical stimulation as an adjunctive therapy to wound healing is supported in the literature.²⁷⁻³¹ Electrical stimulation is thought to facilitate wound healing by providing electrical current to the injury, causing galvanotaxis, stimulating cells, enhancing blood flow, combating infection, reducing edema, and enhancing autolysis. Indications include chronic or recalcitrant wounds. Contraindications include untreated osteomyelitis and concurrent use with dressings containing heavy metal ions. Common methods of treatment include high volt direct current with direct or periwound application at a frequency of 80 to 125 Hz, submotor (no contraction seen) intensity of 75 to 200 volts, with 45 to 60 minutes of treatment 3 to 7 days a week, or 30 minutes twice a day. With the direct treatment method of placing an electrode directly in the wound bed and the ground electrode placed in a location to maximize current flow through the wound. Polarity recommendations vary but most recommend placement of the cathode in the wound bed for an infected wound, and placement of the anode in the wound bed when promotion of granulation tissue and epithelialization are desired. The polarity should be changed when a plateau of healing is observed.³²

Negative Pressure Wound Therapy

Use of negative pressure wound therapy (NPWT), also known as vacuum-assisted closure, may enhance wound healing through topical subatmospheric pressure applied to the wound bed. An open-cell foam cut precisely to the size of the wound is placed within the wound bed and an evacuation tube with side ports is embedded within the foam with the free end of the tubing exiting at one end. The foam contact layer may be lightly moistened with saline to improve comfort. A skin sealant may be applied to the periwound area and a hydrocolloid or sheet hydrogel dressing may be placed where the evacuation tube exits the foam to protect the underlying tissue. The film drape is then applied over the foam and wound with enough overlap in the periwound area to ensure an airtight seal.³³ The evacuation tube with a clamp is attached to an adjustable negative pressure pump and canister. The amount of negative pressure typically ranges from 50 to 175 mm Hg and may be set for continuous or intermittent vacuum.³⁴ Although various pressures are recommended based on wound etiology, the most common pressure is approximately 125 mm Hg because this pressure has been associated with maximum increase in blood flow. Lower pressures may be used for chronic wounds or over skin grafts and higher pressures may be used for acute wounds or wounds with large cavities.³⁵

The beneficial effects of NPWT appear to result from increased blood and lymphatic flow; enhanced tissue perfusion, exudate management, release of growth factors, cell proliferation; and reduced edema. Contraindications

include use on dry wounds, necrotic wounds, in body cavities, over malignancies, or in the presence of exposed blood vessels. Use is indicated for acute, chronic, and traumatic wounds; partial thickness burns; pressure ulcers; and neuropathic wounds.³⁶

Holistic Treatment

Throughout this chapter, the primary emphasis has been on local wound care. However, the clinician must recognize and attend to global factors that may affect wound healing, such as nutrition of the patient, adherence of all staff to exercising excellent wound management procedures at all times, and the capabilities of the at-home caregivers. It is critical that the patient receive sufficient amounts of water, protein, fats, and carbohydrates to support wound healing. Maintenance of an appropriate fluid balance is essential for all cellular and systemic functions. Protein deficiency predisposes the patient to altered osmotic pressures, resulting in interstitial edema and restriction of blood flow, which may decrease the body's oxygen diffusion into the affected area. In the presence of inadequate protein and caloric intake, the nitrogen balance of the body is disturbed, resulting in the use of stored proteins for energy, leaving little for wound repair processes. Carbohydrates, particularly glucose, have a protein-sparing effect, allowing the body to use fats and carbohydrates for energy. Fats carry fat-soluble vitamins, help with thermoregulation, and, in the form of free fatty acids, compose important components of cell membranes vital to the synthesis of new cells.

Provision of a proper wound healing environment at the local and systemic level leads to effective and faster wound healing. Through proper wound cleansing, debridement, dressing management, and nutritional support, the body should have the elements necessary to provoke an appropriate inflammatory, proliferative, and maturation response that results in healed wounds within the typical time frames.

Conclusion

Rehabilitation patients frequently have wounds, including surgical wounds and wounds resulting from nonsurgical trauma. A good understanding of wound healing and the care of wounds is necessary to speed the rehabilitation process and identify abnormal wound healing. This chapter summarizes the wound healing phases and discusses factors that may delay the tissue restoration process. Modalities, dressings, and alternative treatments to speed the healing process are discussed. Further reading in this specialty can be found in these resources.³⁷⁻⁴¹

REFERENCES

1. Sepahl GC, Woodward SC: Repair, regeneration, and fibrosis. In Rubin E, editor: *Rubin's pathology: clinicopathologic foundations in medicine*, ed 4. Philadelphia, 2005, Lippincott Williams & Wilkins.

2. Doherty D, Haslett C, Tonnesen M et al: Human monocyte adherence: a primary effect of chemotactic factors on the monocyte to stimulate adherence to human endothelium. *J Immunol* 138(6):1762-1771, 1987.
3. Sussman C, Bates-Jensen BM: Wound healing physiology: acute and chronic. In Bates-Jensen B, Seaman S, editors: *Wound care: a collaborative practice manual for health professionals*, ed 3. Philadelphia, 2007, Lippincott Williams & Wilkins.
4. Myers BA: Wound healing. In Myers BA, editor: *Wound management: principles and practice*, ed 2. Englewood Cliffs, NJ, 2008, Pearson Prentice Hall.
5. Murphy HS, Ward A: Inflammation. In Rubin E, editor: *Rubin's pathology: clinicopathologic foundations in medicine*, ed 4. Philadelphia, 2005, Lippincott Williams & Wilkins.
6. Seiler WD, Stahelin HB: Implications for research. *Wounds* 6:101-106, 1994.
7. Kerstein MD: Moist wound healing: the clinical perspective. *Ostomy/Wound Manage* 41(7A):37S-43S, 1995.
8. Sephal GC, Woodward SC: Repair, regeneration, and fibrosis. In Rubin E, editor: *Rubin's pathology: clinicopathologic foundations in medicine*, ed 4. Philadelphia, 2005, Lippincott Williams & Wilkins.
9. Payne DN, Babb JR, Bradley CR: An evaluation of the suitability of the European suspension test to reflect in vitro activity of antiseptics against clinically significant organisms. *Lett Appl Micro* 28:7-12, 1999.
10. Drosou A, Falabella A, Kirsner RS: Antiseptics on wounds: an area of controversy. *Wounds* 15(5):149-166, 2003.
11. Gilchrist B: Should iodine be reconsidered in wound management? *Eur Tiss Repair Soc J Wound Care* 6(3):148-150, 1997.
12. Burks RI: Povidone-iodine solution in wound treatment. *Phys Ther* 78(2):212-218, 1998.
13. Sibbald R, Orsted H, Schultz G et al: Preparing the wound bed: focus on infection and inflammation. *Ostomy/Wound Manage* 49(11):92-99, 2003.
14. Schultz GS, Barillo DJ, Mazingo DW et al: Wound bed preparation and brief history of TIME. *Int Wound J* 1(1):19-32, 2004.
15. Porth CM: *Pathophysiology: concepts of altered health states*, ed 7. Philadelphia, 2005, Lippincott-Raven.
16. Hedlund CS: Surgery of the integumentary system. In Fossum TW, editor: *Small animal surgery*, ed 3. St. Louis, 2007, Elsevier.
17. Stewart PS, Costerton JW: Antibiotic resistance of bacteria in biofilms. *Lancet* 358(9276):135-138, 2001.
18. Bill TJ, Ratliff CR, Donovan AM et al: Quantitative swab culture versus tissue biopsy: a comparison in chronic wounds. *Ostomy Wound Manage* 47(1):34-37, 2001.
19. Levine NS, Lindberg RB, Mason AD et al: The quantitative swab culture and smear: a quick, simple method for determining the number of viable bacteria in open wounds. *J Trauma* 16(20):89-94, 1976.
20. Dyson D: Analgesia and chemical restraint for the emergent patient. *Vet Clin North Am Small Anim Pract* 38(6):1329-1352, 2008.
21. Mathews KA: Nonsteroidal antiinflammatory analgesics: a review of current practices. *J Vet Emerg Crit Care* 12(2):89-99, 2002.
22. Grubbs BC, Statz CL, Johnson EM et al: Salvage therapy of open, infected surgical wounds: a retrospective review using Techni-Care®. *Surg Infect* 1(2):109-114, 2000.
23. Nakao H, Yamazaki M, Tsuboi R et al: Mixture of sugar and povidone-iodine stimulates wound healing by activating keratinocytes and fibroblast functions. *Arch Dermatol Res* 298(4):175-182, 2006.
24. Mphande A, Killowe C, Phalira S et al: Effects of honey and sugar dressings on wound healing. *J Wound Care* 16(7):317-319, 2007.
25. Hedlund CS: Surgery of the integumentary system. In Fossum TW, editor: *Small animal surgery*, ed 3. St. Louis, 2007, Elsevier.
26. Knieriem M: *Hyperbaric oxygen use in emergency and critical care*. International Veterinary Emergency and Critical Care Symposium, San Antonio, TX, 2007.
27. Huttenlocher A, Horwitz AR: Wound healing with electric potential. *NEJM* 356(3):303-304, 2007.
28. Bayat M, Asgari-Moghadam Z, Maroufi M et al: Experimental wound healing using microamperage electrical stimulation in rabbits. *J Rehabil Res Dev* 43(2):219-226, 2006.
29. Bai H, McCaig CD, Forrester JV et al: DC electric fields induce distinct preangiogenic responses in microvascular and macrovascular cells. *Arterioscler Thromb Vasc Biol* 24(7):1234-1239, 2004.
30. Houghton PE, Kincaid CB, Lovell M et al.: Effect of electrical stimulation on chronic leg ulcer size and appearance. *Phys Ther* 83(1):17-28, 2003.
31. Gardner S, Frantz RA, Schmidt F: Effect of electrical stimulation on chronic wound healing: A meta-analysis. *Wound Repair Reg* 7:495-503, 1999.
32. Myers B: *Wound management: principles and practice*, ed 2. Upper Saddle River, NJ, 2008, Prentice Hall.
33. Gupta S, Cho: A literature review of negative pressure wound therapy. *Ostomy Wound Manage* 50(11A Suppl):2S-4S, 2004.
34. Kloth LC: Five questions—and answers—about negative pressure wound therapy. *Adv Skin Wound Care* 15(5):226-229, 2002.
35. Lanz OI: *Vacuum assisted closure: current veterinary applications*. In NAVC Conference Veterinary Proceedings: Small Animal and Exotics, Book 2. Orlando, FL, 2007.
36. Armstrong DG, Lavery LA, Diabetic Foot Study C: Negative pressure wound therapy after partial diabetic foot amputation: a multicentre, randomized controlled trial. *Lancet* 366(9498):1704-1710, 2005.
37. Myers BA: Factors affecting wound healing. In Myers BA, editor: *Wound management: principles and practice*, ed 2. Englewood Cliffs, NJ, 2008, Pearson Prentice Hall.
38. Halling K: Wounds and open fractures. In Mathews KA, editor: *Veterinary emergency and critical care manual*, ed 2. New York, 2007, Lifelearn.
39. Swaim SF, Henderson RA: *Small animal wound management*, Philadelphia, 1990, Lea & Febiger.
40. Dernell WS: Initial wound management. *Vet Clin North Am Small Anim Pract* 36(4):714-773, 2006.
41. Garzotto CK: Wound management. In Silverstein DC, Hopper K, editors: *Small animal critical care medicine*. St. Louis, 2008, Saunders.



37 Physical Rehabilitation for the Critically Injured Veterinary Patient

Anne Marie Manning and Zoran Vrbanc

The critically injured veterinary patient can present a daunting challenge to the clinician trying to provide physical rehabilitation. Types of injuries sustained may range from single or multiple orthopedic injuries; soft tissue trauma to the chest, abdomen, or head; or a combination of these injuries. Burn injury or smoke inhalation can cause severe metabolic and cardiorespiratory dysfunction. The common thread in all of these patients is a debilitating injury that renders them partially or completely immobile for prolonged periods. Although bedrest is important to the recovery from critical injuries, prolonged inactivity can have deleterious effects on both injured and healthy body systems. The goals of providing physical rehabilitation to critically injured patients are to prevent complications stemming from injuries, prevent complications secondary to prolonged inactivity, maintain body condition, improve patient comfort, and hasten recovery. The purpose of this chapter is to discuss the effects of prolonged inactivity on traumatized and healthy body systems and to describe the application of basic physical rehabilitation techniques for specific critical injuries.

The Musculoskeletal System

Rarely do traumatized animals escape injury to the musculoskeletal system. Injuries can include single or multiple fractures of the appendicular or axial skeleton, hemorrhage into skin, muscle, and joint spaces, or injury to nerves supplying these structures. Patients with head, chest, or abdominal trauma may be unable to move because of decreased consciousness or pain. Disuse or injury to the musculoskeletal system may result in muscle weakness and muscle atrophy; skin, muscle, and joint contractures; and eventually may limit an animal's ability to move, groom, eat, and ambulate. With these global implications maintenance of this body system is paramount.

The balance between collagen synthesis and degradation is abnormal in immobile patients. Trauma with bleeding into soft tissue and muscle and subsequent inflammation and degeneration can trigger increased collagen synthesis. Increased collagen synthesis in an inactive patient leads to tighter packing of collagen fibers and contributes to the development of contractures.

Muscle contracture is shortening of the muscle owing to either intrinsic or extrinsic causes. Intrinsic changes are structural and can result from inflammatory or traumatic processes. Trauma, inflammation, ischemia, and hemorrhage in injured patients can restructure muscle tissue components and result in fibrosis and the development of adhesions. Following hemorrhage into a muscle, fibrin deposition occurs. Within 2 to 3 days fibrin is replaced with reticular fibers that form a loose connective tissue network. If the affected muscle is kept immobile, the network becomes more dense and resistant to stretch.¹

Extrinsic muscle contracture is secondary to neurologic abnormalities or mechanical factors and is commonly associated with prolonged immobilization. Causes of extrinsic muscle contracture include paralytic or spastic muscle conditions. Paralyzed muscles cannot provide adequate resistance to opposing muscles across a joint. Eventually the antagonistic muscle will shorten. Carpal contracture in brachial plexus injuries illustrates this type of extrinsic contracture. To prevent contracture of the carpus in this situation, stretch must be applied to the normal muscle. Spastic muscle, which may exist in the pelvic limb musculature following trauma to the spinal cord between T3-L3, results from imbalance of muscle control. Increased muscle tone reduces the resting length of the spastic muscle and results in abnormal joint positioning. Treatment is aimed at stretching the abnormal muscle. Joint contracture is defined as an inability to move a joint through its full range of motion (ROM). Conditions that contribute to joint contracture include absence of mobility at the joint, immobilization of a joint in an inappropriate position, joint pain, paralysis, and muscle damage.

The position in which limbs are immobilized also contributes to the development of contractures. Muscle fibers lose up to 40% of the length of sarcomeres when immobilized in a shortened position. Therefore whenever possible joints should be kept in a neutral position to keep muscle fibers at equal length and tension to minimize contracture.² Additional factors that affect the rate of contracture development include the precipitating cause, duration and degree of immobilization, and preexisting joint restrictions. Edema, ischemia, and bleeding accelerate the development of muscle and joint contracture. Immobile, injured patients

Table 37-1 Treatment of Musculoskeletal Injuries

Technique	Benefits	Frequency
Cryotherapy	Reduces inflammation, swelling, and pain	Every 4-6 hr (after initial 24-72 hr)
Superficial or deep heating	Improves circulation, reduces muscle spasm, provides pain relief	Every 4-6 hr (after acute inflam. has subsided)
Massage	Increases circulation to paralyzed musculature, reduces/breaks down adhesions, provides pain relief, reduces edema	4-6 times/day
Range-of-Motion (ROM) Exercises		
<i>Passive</i>	Maintains existing mobility of joints and soft tissue, minimizes contracture formation, prevents loss of ROM, maintains muscle elasticity, improves circulation, reduces edema	4-6 times/day
<i>Active</i>	Same as passive ROM exercise plus maintains physiologic muscle elasticity, maintains bone integrity, and may increase muscle strength	4-6 times/day
Splinting and casting	Corrects severe muscle and joint contracture by providing prolonged stretch to muscle and joint structures	

therefore require efforts directed at prevention and treatment of contracture development following soft tissue musculoskeletal injuries (Table 37-1).

Respiratory System

The respiratory complications of prolonged immobility and those resulting from primary injuries are numerous and potentially catastrophic in critically injured patients. Injuries to the head, chest, and abdomen often result in reduced respiratory function because of pain, inability to move, altered consciousness, or damage to thoracic structures. Impaired consciousness from head injury may lead to altered breathing patterns, diminished ability to cough, and inability to change body position. Rib fractures, pneumothorax, pulmonary contusions, and painful soft tissue injuries to the chest and abdomen may result in a restrictive breathing pattern characterized by shallow, rapid ventilation. This type of breathing pattern causes a reduction in tidal volume, functional residual capacity, and lung compliance. Altered respiratory patterns in conjunction with immobility may lead to atelectasis, accumulation of respiratory secretions, and pneumonia. It is important for the clinician to note any injuries that may compromise respiratory function and devise a physical rehabilitation plan to address those concerns. Chest physical rehabilitation is a term that describes the use of techniques aimed at improving lung volumes or facilitating the removal of airway secretions.³ The goals of chest physical rehabilitation are to maintain bronchial hygiene, eliminate secretions from the airways, re-expand atelectatic lung segments, improve oxygenation, and reduce the incidence of pneumonia

Table 37-2 Goals of Chest Physical Rehabilitation

Technique	Goals
Positioning	Reduce risk of atelectasis, improve ventilation/perfusion, improve gas exchange, reduce risk of pressure sores, minimize muscle and joint stiffness
Cough	Remove secretions from trachea to fourth-generation bronchi
Postural drainage	Increase mobilization and elimination of secretions from upper airways
Percussion	Dislodge bronchial secretions
Vibration	Move secretions into larger airways

(Table 37-2). These goals can be achieved through techniques such as stimulating the animal to cough, frequent repositioning, postural drainage, percussion, vibration, and exercise. Each of these techniques is reviewed in detail.

Cough

Coughing is the most important defense mechanism to eliminate retained secretions. A cough can eliminate secretions from the trachea to the level of the fourth-generation segmental bronchi.⁴ An effective cough is initiated by a deep inspiration, followed by closure of the glottis and contraction of the chest wall and abdominal musculature

to generate high intrathoracic pressure. After high intrathoracic pressure is attained, the glottis opens and is followed by rapid expulsion of air during exhalation. Critically injured patients may not be able to initiate a cough because of impaired consciousness, pain, weakness, or injuries to the chest or abdominal wall. These animals can be assisted to cough by applying gentle pressure to the trachea at the level of the third tracheal ring. Placing an animal in sternal recumbency may also improve an animal's ability to cough. Animals should be assisted to cough after postural drainage, percussion, and vibration and before changes are made in the patient's position.

Positioning

Injured animals are often unable to move or change position within their cage. Alternating right, sternal, and left lateral recumbency every 4 hours is recommended for immobile patients. Frequent repositioning helps prevent atelectasis and pooling of secretions in dependent lung segments. Because position influences the distribution of blood flow to the lungs, changing body positions alters ventilation/perfusion ($\dot{V}/\dot{Q}$) relationships within the lungs. When a patient is positioned with the compromised lung segment in a dependent position, that lung receives increased blood flow, resulting in increased $\dot{V}/\dot{Q}$ mismatch and impaired gas exchange.⁵ Therefore if pulmonary contusions or pneumonia are present, positioning the patient with the good lung down improves oxygenation by limiting $\dot{V}/\dot{Q}$ mismatch and improving gas exchange. Improved oxygenation and a reduction in the number of patients with postoperative fever have been documented in humans when side-to-side turning is used.⁶

Frequent position changes in conjunction with comfortable bedding also help reduce the risk of pressure sores, edematous limbs, and muscle and joint stiffness. Repositioning is a simple physical rehabilitation technique that should be employed in all immobile patients.

Postural Drainage

Postural drainage is the use of changes in body position to allow gravity to aid in the removal of tracheobronchial secretions from different lung segments. This method of thoracic physical rehabilitation prevents pooling of respiratory secretions in the lungs, enhances drainage of the periphery of the lung, accelerates clearance of mucus, and increases the functional residual capacity of the lungs.⁷ The patient must be positioned so that segmental bronchi are vertical to the affected lung to allow movement of secretions into the larger airways. After secretions reach the mainstem bronchi and trachea, a cough will aid in their removal. Postural drainage is most effective when used in conjunction with percussion and vibration to loosen thick airway secretions.

In human patients 11 postural drainage positions are used to drain 14 lung segments.⁸ Figure 37-1 illustrates

seven postural drainage positions that can be used to drain specific areas in the lungs of the canine patient.⁹ The duration of postural drainage therapy depends on the number of lung segments requiring drainage, the amount of secretions produced, and the patient's tolerance for the procedure. In general 5 to 10 minutes per position is necessary for this therapy to be effective. Because exercise is superior to postural drainage for mobilization of respiratory secretions, ambulatory patients should be encouraged to walk for 5 to 10 minutes, three to four times a day.⁷ Postural drainage is indicated in injured patients with pulmonary contusions, atelectasis, or altered consciousness, and those patients that are immobile secondary to injuries.

Percussion

Chest percussion (coupage) is performed by rhythmically striking the patient's chest wall with cupped hands over the affected lung segments (Figure 37-2). With this technique a mechanical shockwave is applied to the chest wall and transmitted to the lungs to dislodge secretions from bronchial walls. The goal of percussion in combination with vibration is to move secretions from small to larger airways where they can be cleared by a cough or postural drainage techniques. Percussion is performed throughout both inspiration and expiration. The amount of force used is determined by patient tolerance to the procedure and the patient's medical problem. Correct hand form and positioning over the affected lung segment are more important than the amount of force used. Only affected lung segments should be percussed because normal lung does not benefit from percussion. The rate of percussion should be steady and the therapist should proceed in a circular motion over the affected area.¹⁰ Erratic percussion, that is, frequent stops and starts, should be avoided. Generally percussion should be sustained for 3 to 4 minutes followed by vibration. Three to four periods of percussion are generally adequate for each postural drainage position. Percussion is not recommended for those patients that produce little or no sputum with a cough¹¹ as well as patients with exacerbation of chronic obstructive pulmonary disease (COPD).¹²

Vibration

After each period of percussion, vibration (shaking of the chest wall) is performed to move the dislodged secretions toward larger airways (Figure 37-3). Unlike percussion, which is performed during both inspiration and exhalation, vibration is performed only during exhalation. The technique is performed with both hands placed on the same side of the chest over the affected lung segment. With arms straight and locked the therapist simultaneously contracts all of his or her muscles from the hands to the shoulder and vigorously shakes the chest wall. A mechanical vibrator also may be used. Vibration is performed on four to six consecutive breaths following each episode of percussion.

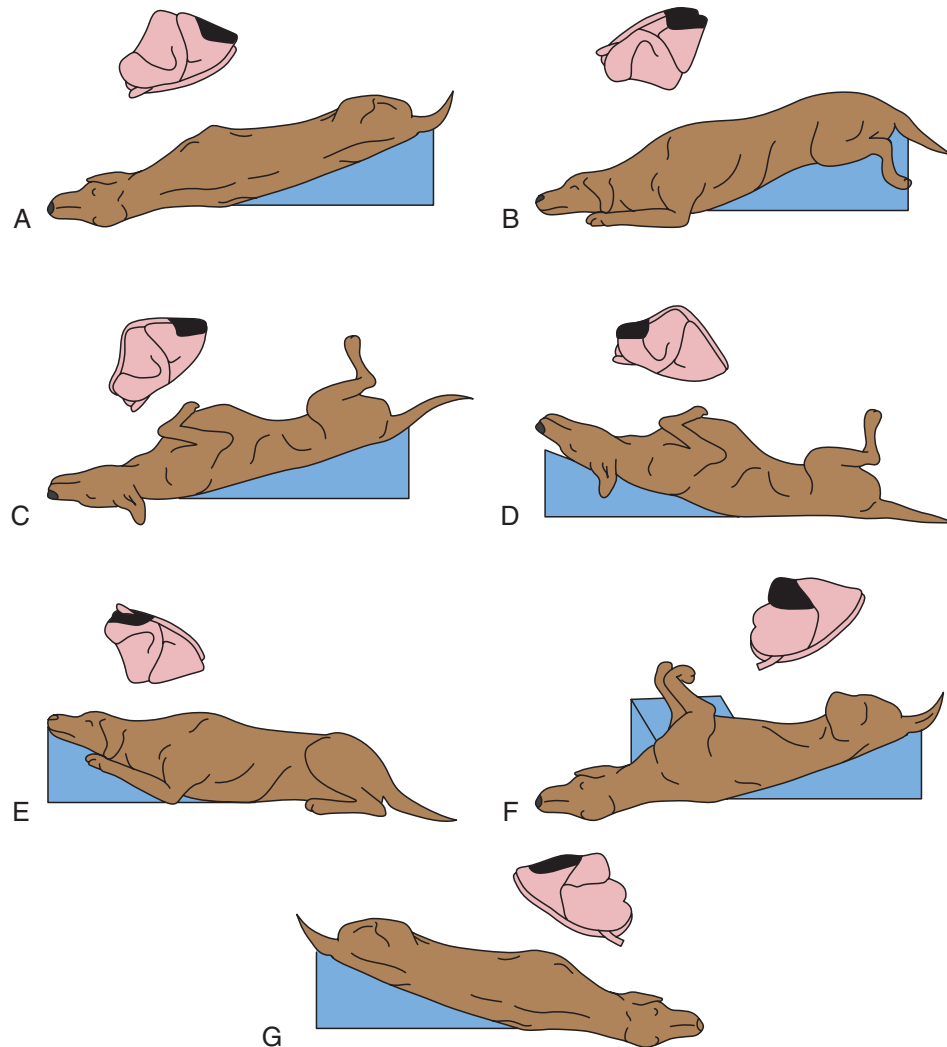


Figure 37-1 Positions for draining various portions of the lungs. **A**, Lateral segment of the left caudal lung lobe. The patient is in left lateral recumbency with the hind end elevated 40 degrees. **B**, Left and right caudal dorsal lung fields. The patient is in sternal recumbency with hind end elevated 40 degrees. **C**, Left and right caudal ventral lung fields. The patient is a dorsal recumbency with the hind end elevated 40 degrees. **D**, Left and right cranial ventral lung fields. The patient is in dorsal recumbency with the front end elevated 40 degrees. **E**, Left and right cranial dorsal lung fields. The patient is in sternal recumbency with the front end elevated 40 degrees. **F**, Right middle lung lobe. The patient is in dorsal recumbency. A pillow has been placed under the left side of the thorax so that the right side is higher than the left side. The hind end is elevated 40 degrees, and the front end is rotated one-quarter turn to the right. **G**, Lateral segment of the right caudal lung lobe. The patient is in left lateral recumbency with the hind end elevated 40 degrees. (Modified from Manning A, Ellis D, Rush J: Physical therapy for critically ill veterinary patients. *J. Chest physical therapy. Compend Contin Educ Pract Vet* 19:680, 1997.)

Any indication for postural drainage is an indication for percussion and vibration.

Contraindications for Postural Drainage, Percussion, and Vibration

Although most patients tolerate chest physical rehabilitation, critically injured patients may be at risk for complications. In human patients postural drainage has been associated with transient decreases in oxygen saturation of arterial blood (SaO_2), and percussion can potentiate cardiac

arrhythmias.^{11,13} Animals with pulmonary contusions often have some degree of hypoxia and should be supplemented with oxygen before and during these techniques. If oxygen desaturation occurs SaO_2 values generally return to pre-treatment values within minutes of position changes and the conclusion of percussion. Animals with chest trauma are at risk for cardiac arrhythmias secondary to traumatic myocarditis. Chest percussion in these animals may precipitate an arrhythmia or cause an existing arrhythmia to worsen. Continuous electrocardiographic monitoring is recommended before and during chest physical rehabilitation.



Figure 37-2 Chest percussion is performed by striking the chest wall with cupped hands over the affected lung segments. The goal is to help break up secretions to larger airways so that they may be expelled by coughing or other means.



Figure 37-3 After percussion, vibration (shaking of the chest wall) is performed to move the dislodged secretions toward larger airways. Vibration is performed only during exhalation. Both hands are placed on the same side of the chest over the affected lung segment. With arms straight and locked, the therapist vigorously shakes the chest wall. Vibration is performed on four to six consecutive breaths following each episode of percussion.

Other injuries in which chest physical rehabilitation is contraindicated are listed in [Box 37-1](#).

Exercise

Exercise is superior to chest physical rehabilitation in reducing atelectasis, mobilizing secretions from the lung periphery, and preventing pooling of secretions in the lungs. A systematic review on human patients suffering from COPD revealed moderate evidence of positive influences of a walking program and chest therapies (except percussion) on patients with COPD exacerbation.¹⁴ Exercise enables deep breathing, which improves ventilation and expiratory flow. Exercise has also been shown to improve sympathetic activity, which increases ciliary beat frequency and decreases mucus viscosity.^{15,16} In addition to the respiratory benefits, early mobilization reduces many of the musculoskeletal complications associated with prolonged immobility.

Box 37-1 Contraindications for Chest Physical Rehabilitation in Critically Injured Patients

- Flail chest or rib fractures
- Pneumothorax
- Platelet count <30,000 platelets/ μ L
- Areas with open wounds
- Subcutaneous emphysema of the neck or thorax
- Pain
- Unstable cardiovascular condition. Use caution with postural drainage techniques in patients with cranial trauma.

Ambulatory patients should be encouraged to stand and walk for short periods of time (5-10 minutes) every 4 to 6 hours. Weak but ambulatory animals can be supported with slings or harnesses. Nonambulatory patients should be assisted to stand for a few minutes every 4 to 6 hours in conjunction with changes in body position. Refer to chapter 29 for examples of exercises that may be used early postoperatively or for patients with extreme weakness and injuries.

Clinical Applications of Physical Rehabilitation for the Critically Injured Patient

Soft Tissue Trauma and Orthopedic Injuries

Trauma to soft tissue structures such as skin and muscle may be confined to a limited area or may be extensive. Extensive bruising, swelling, and pain may result from injury. Initial treatment should consist of immobilization, cryotherapy for 10 to 20 minutes every 4 to 6 hours, and comfortable bedding. If open wounds are evident, protective bandaging should be applied. Animals should be repositioned at least every 4 to 6 hours to prevent dependent edema from forming in the limbs and to prevent lung atelectasis. If the extremities are injured the involved limbs should be elevated above the level of the heart to promote drainage and reduce edema. Bandaging the limb can prevent or reduce edema by compressing lymphatics and blood vessels to limit fluid accumulation in the interstitium. Bandages of cast padding and reusable elastic bandage material should be applied beginning with the distalmost aspect of the extremity.⁹ Bandages may need to be changed several times daily to allow inspection of the area and for physical rehabilitation.

After acute inflammation has resolved, warm compresses or warm water baths may be used. If the animal is capable, restricted exercise is encouraged beginning 48 hours after the initial insult. If spinal cord or orthopedic injuries prevent the patient from ambulation but permit movement of the limb, gentle passive ROM exercises

should be initiated. The limb and all joints should be cycled through their full ROM 10 times, four to six times daily. Range of motion exercises may be performed in conjunction with massage therapy. If fractures are present, ROM exercises should be withheld until appropriate fixation of the fracture. After surgical stabilization of fractures, cryotherapy should be applied to reduce inflammation, pain, and muscle spasms. Following resolution of the acute inflammatory period, application of warm compresses may be beneficial to help relieve pain and muscle spasms. Passive range of motion of the joints above and below the fracture site and massage therapy of the soft tissues are usually initiated 1 to 4 days after surgery if the patient permits.¹⁷ Physical rehabilitation techniques to prevent contracture formation include encouraging ambulation, active and passive range of motion exercises, and stretching affected joints and muscles. Massage may be used to break down adhesions, reduce muscle spasm, and limit edema. If severe contractures have already formed, ROM exercises can be combined with serial casting or splinting. The use of casts or splints allows joints and muscles to be stretched for prolonged periods of time. Moderate contractures may be treated by splinting the area for 20 to 30 minutes, four times daily, in conjunction with ROM exercises and massage. Severe contractures may require placement of a cast or splint for 2 to 3 days. At the end of this period, the area is re-examined, and another cast or splint is placed as necessary to achieve further gains in motion. Splinting and bracing are described in detail in chapter 17.

Burn and Smoke Inhalation Injuries

Severe burn injury and smoke inhalation injury may occur as a result of thermal, radiation, chemical, or electrical injury. Burn injury in animals can lead to severe metabolic, cardiovascular, and pulmonary disorders. The burn classification system classifies burns by the amount of total body surface area (TBSA) involved and by increasing depth: superficial, superficial partial-thickness, deep partial-thickness, and full-thickness.^{18,19} Although burn injuries have not been well described in the veterinary literature, rapid cooling of burns within 30 minutes of thermal injury seems to be beneficial in preventing tissue damage, reducing edema formation, and increasing the speed of re-epithelialization.²⁰ Current recommended treatment includes cooling the wounds with cold tap water (15°C [59°F]) for 20 minutes.²¹ Ice-cold water must be avoided because of the risk of hypothermia.

Smoke inhalation injuries are not common in veterinary practice, but can occur. Smoke contains toxic substances that can cause, apart from respiratory disorders, neurologic signs in animals after inhalation. One of the most important considerations is carbon monoxide (CO), a product of combustion. The ability to bind to hemoglobin multiple times greater than oxygen, results in cellular hypoxia.²² The use of hyperbaric oxygen therapy has been reported

in some articles as controversial, because although hyperbaric oxygen reduces the half-life of carboxyhemoglobin in humans, there are conflicting reports of its efficiency of preventing late sequelae.²³

Heat Stroke

Heat exhaustion and heat stroke are the most common heat-related diseases involving dysfunction or failure of temperature control in the body. They usually occur when the body is subjected to elevated ambient temperature and during strenuous exercise. When body hyperthermia lasts for a prolonged period of time, neurologic abnormalities such as hallucinations, seizures, and coma can occur, with the further effects of organ failure, disseminated intravascular coagulation, permanent brain damage, and death.²⁴

The extent of damage to the body depends on the duration and level of hyperthermia. The primary treatment of an animal suffering from heat stroke is to reduce the body temperature immediately. Cooling methods include external and internal approaches. External methods use the convection and conduction abilities of cold, whereas internal methods are more invasive, including irrigation lavage or intravascular catheters with cool intravenous fluid administration. Some of the described external methods can be practical, such as immersion in a cold water bath or surface application of ice packs. White et al.²⁵ describe the use of evaporative cooling by spraying the dogs with tap water (15°C) (59°F) and compared the results with a peritoneal lavage group. The evaporative technique was as rapid as the lavage group.

Head Trauma

Patients that have sustained head trauma may have impaired consciousness and usually benefit from both chest and musculoskeletal physical rehabilitation. Depressed consciousness may cause abnormal breathing patterns such as hyperventilation, hypoventilation, irregular respiration, or a Cheyne-Stokes breathing pattern (apnea alternating with hyperpnea). Abnormal breathing patterns may lead to reduced tidal volume, functional residual capacity, and lung compliance, allowing respiratory secretions to accumulate. Comatose or stuporous animals are unable to mobilize secretions because of depressed cough and uncoordinated swallowing mechanisms. Atelectasis caused by immobility contributes to respiratory complications. Respiratory physical rehabilitation should be initiated in these patients as soon as they are stable. The therapist should assist the patient to cough and use postural drainage, percussion, and vibration techniques within 48 hours of presentation or as the animal's condition permits. Care should be taken when placing these patients in a head-down postural drainage position. If increased intracranial pressure or ongoing hemorrhage is suspected, head-down positions should be avoided.

Comatose or stuporous patients often exhibit abnormal trunk and limb posture. The limbs may be held in rigid

extension or flexion or may be limp. As soon as the patient's condition allows, musculoskeletal physical rehabilitation should be initiated to prevent muscle and joint contracture. A combination of frequent position changes, massage, and passive ROM exercises should be employed.

Pulmonary Contusions

Chest physical rehabilitation is frequently used in human patients with pulmonary contusions. In patients with traumatic chest injuries, chest physical rehabilitation is believed to exert its greatest effect on the smaller airways rather than the larger airways.^{26,27} Before initiating chest physical rehabilitation, chest radiographs should be taken and evaluated for the presence of pneumothorax, flail chest, or rib fractures that would preclude percussion and vibration. Chest radiographs should also be used to identify contused lung segments so that the therapist may choose the proper postural drainage positions and apply percussion and vibration over the appropriate area.

Conclusion

Rehabilitation should begin as soon as the critically ill patient is stable, and before the onset of complications associated with prolonged hospitalization and immobilization. A proactive approach to rehabilitation requires less time and effort and reaps greater rewards than one that is in response to developing problems. Most patients experience benefits from even simple rehabilitation therapies such as massage, thermotherapy, PROM, and controlled exercise regimens that involve a minimum amount of time on the owner's or therapist's part.

REFERENCES

- Halar EM, Bell KR: Contractures and other deleterious effects of immobility. In Delisa JA, Gans BM, Bockenek WL, editors: *Rehabilitation medicine: principles and practice*, Philadelphia, 1988, Lippincott.
- Spector SA et al: Architectural alterations of rat hind limb skeletal muscle immobilized at different lengths, *Exp Neurol* 76:94-110, 1982.
- Anderson DM, Keith J, Novak P et al: *Mosby's medical, nursing & allied health, dictionary*, ed 6, St. Louis, 2002, Mosby.
- Smaldone GC, Messina MS: Flow limitation, cough, and patterns of aerosol deposition in humans, *J Appl Physiol* 59:515-520, 1989.
- Wheeler SL: Care of respiratory patients. In Slatter D, editor: *Textbook of small animal surgery*, ed 2, Philadelphia, 1993, Saunders.
- Gibb KA, Carden DL: Atelectasis, *Emerg Med Clin North Am* 2:371-378, 1983.
- Ciesla ND: Chest physical therapy for patients in the intensive care unit, *Phys Ther* 76:609-625, 1996.
- Frownfelter D: Postural drainage. In Frownfelter D, editor: *Chest physical therapy and pulmonary rehabilitation*, St. Louis, 1987, Mosby.
- Manning AM, Ellis DR, Rush JE: Physical therapy for critically ill veterinary patients. I. Chest physical therapy, *Comp Cont Educ Pract Vet* 19:675-689, 1997.
- Percussion and vibration. In Frownfelter D, editor: *Chest physical therapy and pulmonary rehabilitation*, St. Louis, 1987, Mosby.
- Connors AF et al: Chest physical therapy: the immediate effect on oxygenation in acutely ill patients, *Chest* 78:559-564, 1980.
- Snow V, Lascher S, Mottur-Pilson C et al: The evidence base for management of acute exacerbations of COPD: clinical practice guideline Part 1, *Chest* 119(4):1185-1189, 2001.
- Hammon WE, Connors AF, McCaffree DR: Cardiac arrhythmias during postural drainage and chest percussion of critically ill patients, *Chest* 102:1836-1841, 1992.
- Tang CY, Taylor NF, Blackstock FC: Chest physiotherapy for patients admitted to hospital with an acute exacerbation of chronic obstructive pulmonary disease (COPD): a systemic review, *Physiotherapy* 96:1-13, 2010.
- Baldwin DR et al: Effect of addition of exercise to chest physiotherapy on sputum expectoration and lung function in adults with cystic fibrosis, *Respir Med* 88:49-53, 1994.
- Oldenburg FA et al: Effects of postural drainage, exercise, and cough on mucous clearance in chronic bronchitis, *Am Rev Respir Dis* 120:739-745, 1979.
- Hodges CC, Palmer RH: Postoperative physical therapy. In Harari J, editor: *Surgical complications and wound healing in the small animal patient*, Philadelphia, 1993, Saunders.
- Saxon WD, Kirby R: Treatment of acute burn injury and smoke inhalation. In: Kirk RW, Bonagura JD, editors: *Kirk's veterinary therapy XI*. Philadelphia, 1992, Saunders.
- Bishop JF: Burn wound assessment and surgical management, *Crit Care Nurs Clin North Am* 16(1):145-177, 2004.
- Vaughn L et al: Severe burn injury, burn shock, and smoke inhalation injury in small animals. Part 2: Diagnosis, therapy, complications, and prognosis, *J Vet Emerg Crit Care* 22(2): 187-200, 2012. doi: 10.1111/j.1476-4431.2012.00728.x
- Cuttle L, Kempf M, Kravchuk O et al: The optimal temperature of first aid treatment for partial thickness burn injuries, *Wound Repair Regen* 16(5):626-634, 2008.
- Beasley VR: Smoke inhalation, *Vet Clin North Am Sm Anim Pract* 20(2):545-556, 1990.
- Mariani CL: Full recovery following delayed neurologic signs after smoke inhalation in a dog, *J Vet Emerg Crit Care* 13(4):235-239, 2003.
- Diller KR, Zhu L: Hypothermia therapy for brain injury, *Annu Rev Biomed Eng* 11:135-162, 2009.
- White JD, Kamath R, Nucci R, Johnson C, Shepherd S: Evaporation versus iced peritoneal lavage treatment of heat-stroke: comparative efficacy in a canine model. *Am J Emerg Med* 11(1):1-3, 1993.
- MacKenzie CF, Shin B, Hadi F, Imle PC: Changes in total lung/thorax compliance following chest physical therapy, *Anesth Analg* 59:207-210, 1980.
- Ciesla ND, Klemic N, Imle PC: Chest physical therapy for the patient with multiple trauma, *Phys Ther* 61:202-205, 1981.



SECTION VII

Business Considerations in Veterinary Rehabilitation

38 Development of a Rehabilitation Facility for Small Animals and Business Considerations

Jim Minick and Debbie Gross Saunders

Why Consider a Rehabilitation Enterprise?

The success of any business lies in its ability to take advantage of current trends and technologies to support its growth. Veterinarians today are looking for new ways to expand their practices, specialties, and income. Historically this has been accomplished through expansion of specialty areas (e.g., oncology, dentistry, ophthalmology). Most of these disciplines require specialized training and, in some cases, very expensive equipment. Rehabilitation for animals is similar to that currently being performed on humans. A person undergoing knee surgery would expect a regimented rehabilitation program for increased mobility and pain management. Of the more than 60,000 veterinarians currently practicing in the United States, only a small, but growing, percentage offer any form of rehabilitation for small animals.

Rehabilitation can be prescribed for:

- Presurgical and postsurgical procedures
- Recumbent patients, such as trauma or neurologic cases
- Increased mobility and strength in geriatric animals
- Pain management of injured and geriatric animals
- Conditioning canine athletes
- Improving quality of life
- Sports specific injuries
- Wound care and management
- Arthritis
- Weight management and control

Two scenarios exist for establishing a rehabilitation practice; a stand-alone facility or a facility associated with a veterinary practice. Both have advantages and disadvantages. A stand-alone rehabilitation facility offers the advantage of focusing on only rehabilitation with a team of veterinarians, physical therapists, technicians, and other support staff. A rehabilitation facility within a veterinary practice may be able to optimize benefits to patients by providing total patient care. Integration with a regular veterinary practice allows one to take advantage of economies of scale, especially when considering the purchase and use

of expensive diagnostic equipment such as radiographic, ultrasound, and other imaging equipment. The ability to take a radiograph at a rehabilitation facility can be more efficient for the client if additional information is required for the therapist. In a stand-alone rehabilitation facility all pertinent information must be forwarded to the practitioner before the initial evaluation. Of course professionalism and sensitivity to professional relationships must be considered in a rehabilitation facility housed in a veterinary clinic.

Competence in the area of rehabilitation medicine is a prerequisite to provide legitimate service. Training is imperative for a successful rehabilitation practice. For individuals who do not choose to pursue training or when trained staffing is an issue, partnerships with other medical practices and staff may be used for the rehabilitation component.

Patient evaluation is the most critical component of the process because therapeutic protocols are established based on problems identified during the evaluation. As with humans, rehabilitation of animals involves numerous types and degrees of care. When the evaluation process is thorough, logical, and based on sound clinical fundamentals, positive results can be expected. By following a structured program the results may be further improved. The evaluation will determine the primary, secondary, and possibly tertiary problems with which the dog presents. After the problems are identified, the goals are established and the respective treatments are outlined. Specific considerations for each animal should be considered as well. For example aquatic therapy (Figures 38-1 and 38-2) may not be applicable to every dog for a variety of reasons.

Creating a Sound Business Strategy

When starting a business there are basic strategies that must be considered. No single individual can possess all the answers; therefore a well-planned business strategy must be developed with a common sense approach to success. Common sense is the knack of seeing things as they are, and doing things as they ought to be done.

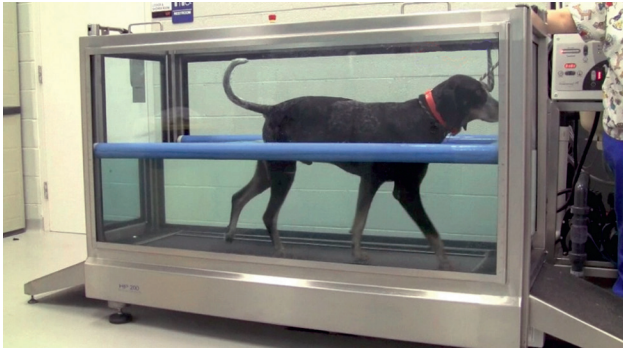


Figure 38-1 Underwater treadmill.



Figure 38-2 Rehabilitation pool.

The road map to a successful rehabilitation practice will result from sound business principles discussed throughout this chapter.

The basic rules for a successful practice may be summarized as follows:

- Know what you want.
- Find out what it takes to get it.
- Act on it and persevere.

As the process progresses, it cannot be stressed enough that the key to success in any worthwhile endeavor is to align yourself with the correct team. It is not difficult to pick the best and brightest individuals who can help you succeed. Canine rehabilitation is a relatively new field and it is important to examine clinics established in different areas of the country. In addition most cities have Small Business Administration (SBA) offices that can be wonderful resources for each of the points discussed in this chapter.

One of the initial steps in establishing a practice is determining whether a physical rehabilitation practice is viable in your community and your location. Some questions to consider include: (1) What is your referral source? (2) Is the location convenient for clients? (3) Are there other canine rehabilitation clinics within a 50-mile radius, and what services do they offer? (4) What are the economic demographics for your location? Even if a rehabilitation clinic is within an existing clinic, some portion of referrals

will undoubtedly be from outside sources. Existing canine rehabilitation clinics within a 50-mile radius is not a negative, but it is advantageous to determine what services they offer and determine how you may compete.

Business Plan

After determining that a rehabilitation practice is something that can enhance a new or existing practice, a written plan must be drafted. Anyone who has considered a new business is always encouraged to have a well-defined written business plan. Until the plan is in writing you are not fully committed to the business entity.

If you already have your own practice you should be familiar with the need for and structure of a good business plan. If not there are many resources that may be consulted to help in this endeavor. Organizations such as the SBA can help to create the plan or, if preferred, numerous books and computer software are available to guide the process. Many existing canine rehabilitation centers have business plans and they may be able to offer advice and guidance.

Legal Entity

A logical plan should start with the choice of a legal entity such as a Corporation, Subchapter S Corporation, or Limited Liability Company (LLC). These entities can provide protection for you and your assets. The state in which you do business, the size of your company, and the potential tax benefits/burdens all influence the entity chosen. Discussions with a lawyer and accountant will help to determine the best entity for each situation. Doing Business As (DBA) may be set up initially under an LLC for a rehabilitation clinic in an existing facility. Advice from legal counsel should be pursued regarding this matter.

Accounting and Legal Support

Two of the most important people to have on any team are a competent business attorney and an accountant with tax experience that specializes in small to medium-sized service businesses. Accounting and legal expertise can save the average business owner thousands of dollars over a 5- to 10-year period. An accountant makes sure bills are paid in a timely fashion, supplies banks with information that is required, such as profits and losses, accounts receivable, and cash flows, and makes sure that a sound accounting system minimizes tax liability. A good attorney can reduce or minimize legal issues related to employees, employment agreements, partnerships, contracts and, to some extent, construction and equipment contracts. If there is an ongoing working relationship with specific individuals, these business associates will help you and your business. These services will be invaluable in making sound business recommendations. The most successful small businesses incorporate tax and legal advice from their inception, making them members of the team and keys to success.

Financing

Financing the venture may be accomplished in a variety of ways. Bank loans are the most obvious avenue to raise funds. Community banks are becoming more creative in their loan practices and may be flexible if there is a good credit history. Alternative sources of funding such as investment partners who provide equity and, in some cases additional business acumen, may present valuable insights. In addition the SBA provides a number of financial assistance programs for small businesses. Credit unions and insurance companies may also provide financing to diversify their portfolios.

Consider segmenting the money needed into startup costs, working capital, equipment financing, and so on. For example an accountant can help determine if a lease is best for acquiring various pieces of equipment needed for the practice. At all costs the temptation to use credit cards for financing, as many small business owners have done, should be avoided. If venture capital funding partners are used, be certain to have an attorney(s) review any repayment agreements. Examine the various costs of rehabilitation clinics and determine the needed equipment. Constructing a needs list and a wants list may be helpful in determining the best financial strategies.

Marketing and Advertising

As the practice and team expand, consider adding someone who can market the practice (whether it is yourself or someone experienced in dealing with customers). One of the most important and sometimes underused focuses for any successful business is marketing. Practically speaking if nobody in the practice has a marketing background, a professional should be sought to assist in marketing efforts. The day-to-day responsibilities of the practice, coupled with the responsibilities as a business owner or manager, leave little time to dabble in marketing. Moreover, with online marketing tools, such as blogs, Web pages, and social media like Twitter, Pinterest, and Facebook, a great deal of time may be required to simply update the electronic marketing presence of the business. In today's society, Facebook and Twitter are both social media sites that are a must for any small business to market their product.

Marketing does not have to be expensive to be done well. Creativity in industries related to animals is welcomed and appreciated. Your marketing strategy should include most, if not all, of the following elements:

- Brochures
- Web site
- Target marketing
- Print advertising
- Cable TV advertising
- Self-promotion through participation in
 - Seminars
 - Local conferences and pet expos
 - Local referral groups
 - Professional associations

Attributes of each of the elements should include:

- Brochures should be colorful, informative, and uncomplicated.
- Web sites should be innovative, easy to navigate, informative, customer friendly, and dynamic. A static (not often changed) Web page can be more harmful than helpful.
- Web blogs can draw countless people to your Web site. Social media like Twitter, Facebook, and Pinterest allow you to contact numerous people with short, logical updates. Any blog visited should provide the opportunity to comment, which should be used to drive traffic to the practice blog or Web site.
- Target marketing to the consumer helps to define the specialty area and narrows the focus of marketing to particular groups such as clubs that participate in agility, tracking, lure coursing, and so on.
- Print advertising can take many forms, including articles written about the facility and the services performed. This media is probably considered more "promotion" than advertising, but the fact that the facility appears in print is the important point for branding and self-promotion.
- Cable TV is effective and relatively low cost in most areas. Cable companies do a terrific job of targeting the message to the area you specify. The key is to create a commercial that is eye catching and memorable. A professional presentation that focuses on the practice's strengths draws attention to the services provided. Promotion of the practice will be assisted by creating a commercial that includes a tag line that is memorable. Repeatability is one of the keys to successful television advertising. Crafting your spot to make it adjustable offers much flexibility and gives the most efficient use of money. Placement of the commercial on specific stations is paramount and is based on demographic information for your area. This information should be thoroughly researched before signing a contract.
- Promotion through continuing education is easy and can cost relatively little. Hosting or participating in educational seminars, hosting an open house, volunteering to speak at dog club events about the benefits of rehabilitation, or participating in local pet expos is a great way to get the word out about your practice and promote the amazing things that rehabilitation can accomplish for a client's animals.
- Advertise your success stories on your Web site, social media, and clinic with many photographs, quotes from owners, and progressions. People love to hear success stories about animals, especially heartwarming stories.
- Word of mouth in the canine world should never be underestimated. Happy clients refer more clients. Each time an existing client refers a client, a personalized letter or card of gratitude should be sent.
- Marketing to referral sources will encompass all of the preceding points. In addition, meetings with referring

veterinarians and regular in-service training programs to the staff should be part of the marketing plan. Packets of brochures, referral pads, and other marketing materials such as magnets and pens should be distributed to potential referral sources. Computer software programs may be used to track referral sources and the number of referrals from each clinic. Communication with the referring veterinarians and clinics through evaluation letters, progress notes, and discharge summaries will enhance the relationship.

- Be certain to obtain written permission from the owner before posting any pictures of their animal or information about the patient or owner.

Suppliers

Competitive bidding from specific suppliers and manufacturers and/or distributors can be invaluable in reducing purchasing costs. The 80/20 rule can be applied when it comes to suppliers. This philosophy gives a favorite vendor the majority of product sales, but keeps him or her competitive because competitors are always trying to gain more market share. Common products sold in rehabilitation clinics include harnesses, slings, fitness equipment, supplements, and educational DVDs. There is also a market for prescription foods, especially those related to rehabilitation and weight loss.

Overhead and Burden

When evaluating the financial implementation of a rehabilitation center, overhead and burden must be considered. Overhead is the cost of doing business and is generally composed of rent/mortgage for the space, utilities, equipment, and staff costs. Overhead costs often overlooked in a rehabilitation setting include increased electricity, dehumidification, water costs for aquatic therapy, cleaning costs for the center, insurance coverage for the clinic inclusive of the aquatic therapy, liability coverage, and workers' compensation for staff members. Burden, specifically labor burden, refers to the benefits that are paid to employees. It is critical to have the correct personnel in place who are well trained and capable of providing outstanding customer service. Most clients of rehabilitation services are repeat customers because of the nature of ongoing care. In the case of geriatric animals, there is a long-term relationship, and clients appreciate the continued interface with the therapist. Trained personnel who have completed an educational rehabilitation program, such as veterinary technicians or physical therapist assistants, can command salaries that range from \$35,000 to \$50,000 annually, depending upon experience and skill sets. Physical therapists trained in canine rehabilitation may command salaries between \$50,000 and \$100,000 annually.

Billing rates for services provided to the customer can range from a low of \$50 per hour to \$75 or more per hour. The rates vary depending on the facility and location. Additional fees can be recovered through:

- Expertise of the DVM
- Evaluation for illness or other rehabilitation services
- Pool/treadmill/manual manipulation
- Specialized equipment, such as therapeutic laser
- Specialized treatments, such as acupuncture or chiropractic manipulation

Information Technology Systems

Information technology (IT) has become increasingly important in running an efficient business. It allows for the integration of many functions, eliminating the need to do or write things more than once. In all likelihood the investment in IT will represent a larger percentage of the budget than might originally be anticipated. Fortunately this new technology can ultimately help save time and money. New technologies are developed regularly for client billing, communication, documentation, correspondence, medical records, video conferencing, and so on. A myriad of practice management software programs can seamlessly integrate accounting and operational functions. For example it is possible to attach radiographs and other image files directly to client files where they can be viewed from any designated computer. Consultations with specialists can be efficiently and conveniently managed by sending files electronically to anyone in the world. The investment in equipment, which is expandable and secure, offers immeasurable returns by offering the business owner tools that can help operate the business and communicate with clientele. In fact, many programs allow client interaction via the Internet if desired.

Practice management software allows reminders to be sent to clients to increase compliance in a rehabilitation practice, appointment scheduling, patient information updates online, and client surveys based on specific services.

Roles and Integration of Staff Members in a Rehabilitation Facility

Most state practice acts stipulate that a licensed veterinarian must oversee and regulate the rehabilitation facility. Because rehabilitation naturally encompasses modalities related to physical therapy, it is not only important to understand the veterinary practice acts in your state, but additionally, an understanding of the physical therapy practice acts is critical to the successful administration of a rehabilitation practice.

The unique nature of animal rehabilitation has made it possible for physical therapists, initially trained to work with humans, to now work with animals. Unheard of only 20 years ago, it is now practical for both a veterinarian and physical therapist to work together to create a comprehensive program that addresses both the biologic and physiologic well-being of animals. Ideally the rehabilitation team should be led by a veterinarian who is responsible for the overall health of the animals, and a physical therapist who develops the therapeutic programs and administers treatment to the animals.

To be practical, the success of any business depends on the volume of service it is able to offer. To that end, veterinary technicians should be employed to help provide hands-on care for the animals in addition to regular client interaction and education. This interaction enables the client to be more proactive in the animal's home environment through education and training, and assists the animal's rehabilitation (e.g., at-home exercises), which provides continued therapy designed to further the results of in-house rehabilitation.

After a treatment regime has been determined the veterinary technician can perform many of the day-to-day treatments of the animal. Having a veterinary technician perform many of the treatments allows the veterinarian and physical therapist to perform other professional duties, including consulting on additional cases. Of course it is vital to have frequent communication among the veterinary technician, veterinarian, and physical therapist.

Furthermore veterinary assistants can be invaluable in helping veterinary technicians with some tasks, such as supervising routine treadmill sessions or drying patients after aquatic treatments. They can also assist with notations in the medical record during patient evaluations with the veterinarian or physical therapist. This promotes an efficient workflow, and perhaps more accurate recording of notes. Technology also currently exists that allows dictation of notes directly to the computer.

For a business owner it is natural to provide training to team members according to the needs of the practice. In most businesses this is practical and, indeed, necessary for processes to be performed correctly. Continuing education is also critical for employees to obtain new information and to incorporate new methods and practices. It is also beneficial from the standpoint of running a business to encourage and support personnel in their desire for advancement in the field. Therefore it is imperative for the owner of a successful rehabilitation facility to maintain a provision for continuing education of all employees on the profit & loss statement. Any technician who works directly with patients should receive training in veterinary rehabilitation. Specialized training not only provides personnel with needed information, but it will also be evidence to current and potential clients that the staff are properly trained and enthusiastic about providing the utmost quality of care.

Facility

The size of a rehabilitation facility is generally dictated by the services offered as well as the anticipated patient flow. Like an animal hospital the design of a rehabilitation facility must take into account the flow of patients (how they will move from room to room), the amount of time they are to spend at each station/room, and the number of patients receiving services at any given time. The floor plan for new office space should include an entrance and an exit so dogs have minimal interaction with one another. Half walls offer views of the facility but do not allow visual dog interactions. Successful rehabilitation practices ideally see

multiple patients at the same time, and the space should allow this through half walls, private rooms, aquatic rooms, and exercise rooms. Modalities and manual therapies are best provided in a quiet section, and aquatics and therapeutic exercises are best provided in their individual sections. The flow of the patients must be considered for both check in and check out.

When evaluating the potential case load consider rehabilitation clients as well as weight loss and conditioning programs. Programs for conditioning and weight loss may consist of aquatic therapy and therapeutic exercise only. These programs may be very profitable; offering continued wellness programs after weight loss goals are accomplished and skilled rehabilitation is completed is another source of revenue. For example a dog that has undergone rehabilitation for a repaired cranial cruciate ligament may be discharged to a conditioning program to maintain the progress.

Facility design is a priority item in the business plan. Design, plan, and functionality determine costs associated with the relocation or construction of a new facility. After the building's structure is completed and the interior walls are set, it becomes extremely costly to make any changes or modifications. Therefore in-depth planning is critical for the success of the project. Before committing to construction plans visit or view photos of other rehabilitation practices to determine what works best for your practice.

Assessment of the Concept and Design

The first priority is to engage a licensed architect who has experience in the design and construction of animal hospitals. Always interview potential business vendors and suppliers and obtain references. Your investigation should include site visits when ever possible to view previously completed projects and discussions regarding the process, cost overruns, punctuality, and responsiveness of the architect and his or her subcontractors. Your questions should cover documentation, scheduling, change orders, materials, and systems used. When evaluating design criteria with the architect, be sure basic needs are captured in the design. Be proactive and clear in your demands for the features required and desired in the facility. It is important to keep in mind that many architects have not created a canine rehabilitation facility and will need to be educated on the dynamics of the facility regarding flooring, noise control, aquatics, office flow, humidification and maintenance.

The approximate space requirement for a rehabilitation practice is 500 to 2500 square feet, depending on the type and amount of equipment and potential client base. This does not include examination/assessment rooms (Figure 38-3) and kennel/run areas (Figure 38-4), which can easily add an additional 2000 square feet. If adding a rehabilitation facility to an existing medical facility, certain in-house areas, such as the examination and housing areas,



Figure 38-3 Patient assessment areas.



Figure 38-4 Kennels/runs.

are usually used in conjunction with the rehabilitation facility.

Proper flooring in the aquatic and exercise areas is paramount, as is noise control if kennel areas are planned. The walls surrounding the aquatic area must be considered with

respect to water. The walls will get wet whether there is an underwater treadmill, a pool, or both. The walls should be comprised of a surface that will endure the constant splashing of water. Tiles, faux tiles, epoxy coating, or specialized paints are options. The surrounding flooring must withstand the water and be safe for the dogs, staff, and owners. HVAC requirements and humidification should also be a prime consideration in the design.

Kennel areas should be as comfortable as possible, and heating and cooling considerations are necessary. If the clinic will be performing long-term boarding for dogs, radiant heat flooring provides a good option for both heating and practicality. Sound-proofing the walls provides a more peaceful environment for the rehabilitation clinic and there are many options available. Ideally the kennel area should be easily accessible to the rehabilitation area, but also far enough away to offer a quiet area.

Architectural Design

Architectural design should encompass the following:

- A clean and simple flow for movement of patients and staff throughout the building. The flow of movement including check in area, treatments, and check out should be considered. Handicapped access and access for dogs in carts must be considered.
- A plan for maintenance and repair of equipment. Be sure that access points to essential components, such as heating, ventilation, and air conditioning (HVAC), electrical, and plumbing are not hidden behind walls or hard-to-move equipment. While in the design process, consider if you want the equipment to be seen by only staff members or the owners. Many designs allow the majority of the equipment to be placed out of view of the public, but allow easy access. For example, the holding tanks and the equipment associated with aquatic therapy are often loud and may need to be in an area away from patient care. A proper design allows easy access to the equipment, but is aesthetically pleasing to the clients and staff.
- Always specify materials that are both cost effective and readily available. Also incorporate materials that can easily be sanitized. Avoid expensive, one-of-a-kind materials that can be hard to replace.
- During the design phase it may be appropriate to engage an HVAC/air purification expert who can address air flow and potential odor issues throughout the facility. Dehumidification is essential in a clinic with aquatic therapy and should be addressed before the construction process.
- Most architects will make recommendations regarding specific manufacturers and products with which they have had experience. Usually this applies to plumbing, electrical, and HVAC components. It can save both time and money in the end if you do your own research with various manufacturers. Technology changes quickly



Figure 38-5 Reception/waiting area.

and animal hospitals built as little as 2 years ago have systems that may already be either outdated or not environmentally friendly.

- Flooring is a major component of the building process and may be a major expense. There are many flooring options available. Floor surrounding the aquatic area must be able to absorb or drain water, provide a non-slippery surface, and be easy to clean. Ergonomically it should possess a friendly surface for staff to stand on for prolonged periods of time. This type of flooring is often more expensive than other flooring. The reception area (Figure 38-5), private offices, staff areas, and other areas may not require expensive flooring. The flooring designs should be implemented before construction.
- New technology should not be avoided if the principles are sound and the equipment has some track record. An example is tankless hot water heaters. These new systems, although more expensive to purchase and install, have a more efficient operating cost along with better longevity and fewer space constraints. They also provide hot water on demand.
- Operating systems necessary for the rehabilitation center include a heating and cooling system.

City and Municipality Permits

Before planning an addition or building a new rehabilitation center, it is essential to begin the process with the legal entity controlling permits. Local regulations may or may not allow the addition of the center, especially with aquatics. A swimming pool is often the issue and may require specialized locks and gates for the center. Stand-alone facilities may face difficulties with individual town regulations because of the nature of the business. Canine rehabilitation does not fall into a specific category and many towns have a difficult time categorizing the clinic. Overnight boarding, noise control, aquatic management, waste, and physical waste management are often issues.

Box 38-1 Considerations for Evaluating Location

- Area needs for services to be provided
- Potential competitors in the area, specifically within a 5-mile radius of your new facility
- Type of facility/services you would provide. For example, a rehabilitation center with 24-hour boarding facility or one with more traditional hours of operation
- Staff needed to support the planned facility.
- Demographics of potential clientele in trade area
- Business growth potential in the area
- Ease of entering and exiting the facility. Consider traffic patterns, ability to make left-hand turns, future highway construction projects that may limit access, and so on.
- Space outside of the building for animal elimination purposes and areas clients may walk dogs, and space for gait analysis and potential for performance equipment such as jumps, obstacles, and space for running. Fencing requirements must be considered in accordance with the municipal ordinances.

Location of a New Facility

When building a new facility, location should be one of the highest priorities in the evaluation process (Box 38-1).

After the area has been established for the site, due diligence is required. This is usually accomplished in one of two ways.

1. Engaging a reputable and experienced real estate broker as your agent to procure a specific site that meets your needs
2. Searching the trade area looking for signs or checking in the local newspaper for available properties

After site selection and preparing a contract to purchase with the sellers, contingency clauses should always be inserted into the contract relative to financing and zoning approval.

Zoning and Contracting

Considered a minor issue by some, improper zoning of the location and the time it will take to make the necessary changes can add 2 to 6 months to the acquisition and building processes, and in some cases, changes in zoning regulations may be denied. The zoning process itself is usually a one-time process and can take two to three face-to-face meetings with city council members and sometimes local residents, who usually reside within 100 feet of the proposed site and want to voice their opinion. The process can cost \$3000 to \$10,000 for attorney fees and services to make changes.

After the purchase contract and zoning issues have been resolved the contractors and builders who will build the facility should be engaged. Individual budgets must be considered in this process. For smaller facilities general

Box 38-2 Design/Build Concept versus Traditional Concept

Design/Build

This concept uses a specific architect and a general contractor working together as a team, or an architectural firm that has a construction arm or affiliation.

Pros

- One source responsibility
- Changes to scope and design are done quickly. Very little interface with construction arm as it deals with change orders.
- Fewer subcontractors are generally involved in the bidding process
- Minimal documentation as it relates to bank draws
- Fewer people are involved in the whole process

Cons

- Can be more expensive than the traditional method because fewer subcontractors compete in the bidding process
- Relationships can become strained if problems develop during the construction cycle.
- Communication with all parties can be challenging and time consuming.
- Problems may arise if subcontractors need to be replaced for nonperformance issues.

Traditional Method (Use of a General Contractor)

This concept requires that an independent general contractor be engaged directly. After the architectural plans are completed and approved, a general contractor is engaged. A contractor who has experience in building animal hospitals is highly recommended. Be cautious in engaging a general contractor who has no experience building health care facilities and may not be familiar with critical codes and regulations.

Most general contractors solicit bids from at least two or more subcontractors for the many different components of the process, which include:

- Site preparation
- Foundation
- Structure (masonry, frame, etc.)
- Plumbing

- Electrical
- HVAC
- Roofing
- Interior framing/dry wall and finishing
- Other

Pros

- This can be more cost effective.
- The general contractor can solicit bids from numerous contractors to get the most competitive price.
- A diversified subcontractor base can lead to better design with cost-effective changes.
- There is good control of the project.
- If subcontractors must be replaced, changes generally can be made quickly.
- Architectural interface is usually minimal unless design issues arise.
- Day-to-day issues are acted on in a timelier manner.
- There is consistent oversight of subcontractors and the project.
- Changes can be made quickly.
- Disputes can be resolved quickly.
- Timelines are more often met.
- Bank draws are certified before submission to the title company.
- Documentation is addressed quickly.
- The scope of changes can be justified or denied quickly.
- The potential time and cost savings can be significant.

Cons

- There is a high reliability on subcontractors instead of the architect.
- There are multiple layers of communication.
- Conflicts may arise if there are problems.
- More extensive paperwork and lien waivers are required for draw payouts to subcontractors.
- Staging and sequencing can become a problem.
- The time element can be increased because of the complexity of construction.

contractors may be able to help with the design of the facility and may eliminate the need for an architect. There are two basic concepts that must be addressed to determine the best situation for your purposes (Box 38-2).

Regardless of which method is chosen management of the entire process can be handled effectively with the use of a third-party owner's representative that has an architectural or construction background. The role of this individual is to oversee the entire project on behalf of the owner. The third-party provider usually does daily on-site visits, reviews change-orders with the owner, and reviews all bank draws for accuracy. Issues that may arise are cleared through this third-party representative. The cost of this service can vary based on the amount of interface required, and the fee is traditionally approximately 2% to 5% of the

construction price. The benefit is usually a 10% to 20% overall saving on the project, including the time factor of making on-site decisions. Many smaller facilities adding on to their existing facilities may find that general contractors familiar with veterinary clinics are often the most beneficial to making additions to the clinic, and usually have architectural support services available to them.

In short the construction process can be complicated and confusing. The ability to rely on construction professionals and the opportunity to have an individual or group who can act on your behalf helps to make the process easier. Finding those individuals can be tricky, but a little research on your part, concentrating on the experience, references, and review of past projects of the candidates, is the key in choosing those resources.

Rehabilitation Equipment

There are two basic ways to acquire rehabilitation equipment, which can be costly (e.g., whirlpools, underwater treadmills, therapeutic laser).

1. Purchase: Generally there are financing options through a bank or other source.
2. Lease/Lease with Option to Buy: This is common practice and a very good option for both tax purposes and acquisition of updatable technologies. This method allows the practice to take full advantage of tax credits and deductions along with the opportunity to exchange older technology with new, state-of-the-art equipment without having to sell or trade the old units.

No matter how well manufactured, all equipment needs maintenance and occasionally breaks down. When selecting vendors be certain to form an ongoing working relationship with companies that provide service of the equipment. Price is not always the most important factor in making the decisions. There is some truth in the saying,

“You get what you pay for,” but that does not mean that the highest-priced item should always be purchased either. The key is the relationship with vendors and the service they can provide should equipment need repair. A non-working unit can cost a great deal in lost revenues when it cannot be used to provide rehabilitation services. When a call is made for service, will assistance be provided in a timely manner? Determine who will service your equipment, especially the aquatic units and where they are located. Some companies only offer one service provider for an entire region of the United States.



Figure 38-6 Treatment mat.

Box 38-3 List of Supplies and Equipment for a Model Rehabilitation Facility

General Patient Management

- Bandage material
- Booties, different sizes
- Harnesses, different sizes
- Slings, different sizes
- Clippers
- Leashes, with rehab facility logo
- Muzzles, different sizes
- Padded table
- Rolling lift system with supporting sling
- Rolling low gurney to transport weak/paralyzed patients
- Mats
- Goniometers
- Gulick II tape measures
- Hi Lo table
- Scale
- Force plate analysis—stance analyzer

Exercise Equipment

- Wobble board
- Quadraped balance board
- Cavaletti poles
- Vertical weave poles
- Land treadmill
- Limb weights
- Stairs/ramp
- Therapy balls/physio rolls, different sizes
- Thera-bands
- Mats
- Mirrors

Modalities

- Cryotherapy—cold packs, ice compression systems
- Hot packs
- Combo neuromuscular electrical stimulation/ultrasound unit

- Electrodes and ultrasound gel
- Extracorporeal shockwave
- Therapeutic laser

Aquatic Therapy

- Underwater treadmill—1 or 2
- Whirlpool (ideal but not necessary)
- Swimming pool (ideal but not necessary)
- Life jackets, different sizes
- Cage dryers
- Towels (consider towels with company logos)
- Harnesses for aquatic therapy
- Cleaning supplies specific to the aquatic systems
- Water treatment—chlorine, bromine, pH, etc.

Practice/Office Equipment

- Freezer/refrigerator
- Microwave
- Camera with video capability
- Computers/color printer
- Potential mobile computer devices for staff documentation, such as iPads
- Office furniture—desk, phone, chairs, file cabinets
- Telephone/facsimile system
- Desk and chairs
- Client seating/office furniture

Office Management

- Washer/dryer
- Wet vacuum
- Dry vacuum
- Cleaning supplies
- Mop bucket
- Tool kit
- First aid kit
- Trash cans

The total cost of equipment for a rehabilitation facility can range from \$10,000 to \$150,000 depending on what is required to provide the desired services. To choose the necessary items, consult with practices or organizations that currently offer rehabilitation services or distribution networks that supply components to similar businesses. A basic list of suggested items such as treatment mats (Figure 38-6) is provided in Box 38-3.

Strategic Alliances

Strategic alliances with manufacturers of equipment and suppliers can allow the practice to run more efficiently and offer an opportunity to try new medications, equipment, and so on. It is generally easier for manufacturers or suppliers to get their products into the market if they have a proven track record, so many are willing to allow a trial of new products (sometimes at a reduced cost) if a testimonial is offered. Most appreciate honest opinions regarding their products so that they can make improvements.

Customer Service

Who are the most important people in your organization? Veterinarians? Veterinary technicians? Physical therapists? The answer is yes, but without committed and caring customer service representatives there will be no long-term business. Customer service begins with the receptionist, who usually answers the phone, and the person who takes the client and his or her pet into the exam room. People do business with people they like! If a customer has a bad experience, how does he or she react? Nine out of ten simply do not return. How is a difficult situation handled? Most people do one of three things: (1) Nothing, (2) hope for the best, or (3) confront the situation head-on. The best response of course is the third choice. Most people appreciate honesty and the opportunity to discuss and hopefully rectify a situation. It may require a phone call or personal interface from the veterinarian or therapist. Personal contact is usually the best way to resolve an issue.

Canine physical rehabilitation allows a bond to be established between clients and practitioners that is not always witnessed or allowed from a time management point of view in other areas of traditional veterinary medicine.

How do we understand what customers want? Ask them. Customer surveys are a great way to receive feedback. Surveys should be concise, short, have no ambiguous questions, and should allow targeted and meaningful feedback to the practice. People are generally willing to

answer a few questions, especially if they do not have to write a dissertation. Immediacy is also important. A call to the customer to inquire about the experience is an invaluable tool. Web surveys are also an inexpensive, data-rich tool that can be used to improve service. In fact, some software can be programmed to send the Web survey automatically after the patient's appointment. Most customers who have a good experience are happy to refer your services if they are asked. But customers who have a bad experience may take every opportunity to tell everyone they know about that experience, whether asked or not. Bad news generally spreads quickly while good news spreads slowly.

How is the staff trained?

- Take a course on the basics of customer service
- Evaluate and improve how things are done on a daily basis
- The do's and don'ts of customer interface
- Decision on attitude (a bad attitude in an organization can spread like wildfire.)

An often overlooked but important detail is appearance. As an example consider that you are about to board an aircraft and you look in the cockpit and see that the pilot is wearing a t-shirt with holes in it, and has uncombed hair and a 3-day beard. Most people would probably get off the aircraft.

Image! Today's world is very image conscious. Make sure that personnel are well-groomed and have a neat appearance. Shirts, polo shirts, tops, and sweaters with the company's logo are beneficial and provide a great image to clients.

Summary

In summary, four basic components must be considered when opening a rehabilitation practice:

- A good business plan
- The right team members
- Marketing and customer service
- Personnel who are properly trained and educated in the science and art of rehabilitation

As a veterinary practitioner one of the last things you may want to think about is the business of running your business. Most people would rather spend their time and efforts concentrating on veterinary medicine and rehabilitation rather than managing an office. With proper planning and a top notch staff you will be able to set up your practice and work with your team to make it successful.



1

APPENDIX

Protocol Development and Protocols

Andrea Henderson, David Levine, Darryl Millis, and Marti Drum

The sample protocols in this appendix are designed to address common conditions in patients not experiencing complications. Individual patient characteristics must be considered when designing an individual patient protocol. These are working protocols that should be modified according to clinical necessity, efficacy, and each patient's progress within the rehabilitation program. These protocols are not to be rigidly adhered to, but serve as guides to allow an effective rehabilitation plan of care. Furthermore, diagnosis and treatment should be provided by trained personnel.

During rehabilitation, particular attention should be paid to any deterioration in the patient's progress. Specifically, attention should be focused on lameness, stiffness, or pain following activity. If the patient appears to have stiffness or increased pain or lameness at any time after a therapy session, the level of activity should be decreased and a slower rate of progression should be instituted. It is very important that dogs be as comfortable and pain-free as possible during the rehabilitation period.

It is important to realize that the dog may not be totally normal even with intensive physical rehabilitation. The patient's condition may preclude return to normal function. However, rehabilitation is key in obtaining and maintaining as much function as possible. If the dog does not appear to be making progress, has any complications, or if there are any questions, the patient should be discussed

with the clinician to determine an appropriate course of action.

Contents

Orthopedic Conditions

Hip

Femoral Head and Neck Osteotomy, [703](#)

Total Hip Replacement, [725](#)

Hip Luxation—Craniodorsal, Closed Reduction, [707](#)

Stifle

CCL Rupture—Extracapsular Repair, [670](#)

CCL Rupture—TPLO/TTA, [674](#)

Patellar Luxation Repair, [713](#)

Distal Femoral Physeal Fractures, [686](#)

Elbow

Elbow Dysplasia—Post-Arthroscopy, [692](#)

General

Osteoarthritis, [711](#)

Neurologic Conditions

Cervical Spondylomyelopathy (Postoperative), [678](#)

Thoracolumbar IVDE (Postoperative), [719](#)

Degenerative Lumbosacral Stenosis—Conservative Management, [689](#)

Fibrocartilaginous Embolic Myelopathy, [698](#)

Degenerative Myelopathy, [683](#)

CCL Rupture—Extracapsular Repair

Phase	Expected Timeframe*	Rehabilitation Clinic Program	Home Program	Outcome Assessment Measures	Criteria for Movement to Next Phase
Preoperative <u>Goals:</u> Prepare for surgery	From diagnosis to time of surgery	Therapeutic Exercises • Passive Range of Motion, 10-20 repetitions, SID-BID • Downhill Treadmill Walking—5-10% decline walking at a slow-to-medium walking speed, 5 min SID-TID • Brief training to aquatic therapy to ensure patient comfort and compliance Modalities • Transcutaneous electrical stimulation for pain relief (15-20 min SID) • NMES for muscle strengthening (15 min SID PRN) • Cryotherapy (15-20 min BID) Medications • Injectable polysulfated glycosaminoglycan agents (<i>discontinue within 24 hours of surgery</i>) • NSAIDs (<i>discontinue non-selective NSAIDs within 48 hours of surgery</i>)	Therapeutic Exercises • Rest in a confined space • Slow, controlled leash-walking only to go outside, no more than five minutes at a time Modalities • Cryotherapy (15-20 min BID) Medications • NSAIDs • Tramadol, Tylenol 3 • NMDA antagonists • Oral chondroprotective agents	• Baseline bilateral stifle ROM (and other joints if applicable)—goniometry • Baseline bilateral thigh circumference • Gait evaluation • Weight	Surgery should be performed as soon as possible after diagnosis to reduce the progression and severity of osteoarthritis as a sequela of motion on an unstable stifle.
<u>Step 1</u> Postoperative to Early Toe- Touching at a Walk <u>Goals:</u> Pain control Swelling and edema control Improve stifle ROM Limit muscle atrophy	Immediate to 48-72 hours postoperatively	Therapeutic Exercises • PROM for stifle flexion/extension of operated limb (10-20 reps BID) • Slow leash walking only to go outside, no more than five minutes at a time Modalities • Gentle massage • Transcutaneous electrical stimulation for pain relief (15-20 min SID-TID) • NMES—quadriceps and biceps femoris muscle groups (15 min SID) • Cryotherapy (15-20 min BID-TID) Therapeutic Laser SID • Cryotherapy (15-20 min BID-TID) Medications • Injectable polysulfated glycosaminoglycan agents • NSAIDs/other analgesics • Oral chondroprotective agents	Inpatient status preferred during this phase If home: Therapeutic Exercises • PROM for stifle flexion/extension of operated limb (10-20 reps BID) • Slow leash walking only to go outside, no more than five minutes at a time Modalities • Gentle massage • Cryotherapy (15-20 min BID-TID) Medications • Injectable polysulfated glycosaminoglycan agents • NSAIDs/other analgesics • Oral chondroprotective agents	• Postoperative bilateral stifle ROM (other joints if applicable)—goniometry • Postoperative bilateral thigh circumference • Response to activity and subjective pain level • Gait evaluation at a walk • Weight	• Consistent toe-touching on operated limb • Adequate resting analgesia • Decreased perioperative swelling and no incisional drainage • ROM 60-80 degrees flexion, 120-135 degrees extension

<u>Step 2</u> Early Weight-Bearing <u>Goals:</u> Continue pain control Reduce swelling Work to return ROM to a more normal state Increase toe-touching weight-bearing	48-72 hours to two weeks postoperatively	Therapeutic Exercises • PROM and flexion/extension stifle stretches of operated limb (10-20 reps) • Slow leash walking only to go outside, no more than five minutes at a time • Slow ground treadmill walking • Gentle weight shifting at a stance and walk • Balance exercises on a soft foam pad or bi-directional balance board to encourage weight-bearing and joint position awareness Modalities • Heat therapy (10-15 min BID-TID, prior to exercises, NOT within 72 hours after surgery or if acute inflammation is present) • Therapeutic laser PRN • Cryotherapy (15-20 min BID) following exercises Medications • Injectable polysulfated glycosaminoglycan agents • NSAIDs/other analgesics PRN • Oral chondroprotective agents	Therapeutic Exercises • PROM and flexion/extension stifle stretches of operated limb (10-20 reps) • Slow leash walking only to go outside, no more than five minutes at a time BID-TID Modalities • Heat therapy (10-15 min BID-TID, prior to exercises, NOT within 72 hours after surgery or if acute inflammation is still present) • Cryotherapy (15-20 min BID) following exercises Medications • Injectable polysulfated glycosaminoglycans • NSAIDs/other analgesics PRN • Oral chondroprotective agents	• Response to activity and subjective pain level • Gait evaluation at walk and trot • Goniometry—Stifle ROM and other joints if applicable • Thigh circumference • Weight • Consistent partial weight-bearing with each stride on operated limb • Minimal pain with light activities • Incisional healing without complications • ROM 60-80 degrees flexion, 120-135 degrees extension
<u>Step 3</u> Consistent Weight-Bearing at a Walk <u>Goals:</u> Achieve normal ROM Improve weight-bearing at a walk and trot	Two to six weeks postoperatively	Therapeutic Exercises • PROM and flexion/extension stifle stretches of operated limb (10-20 reps BID-TID) if normal ROM not yet achieved • Slow consistent leash walking—add 5 min every week if progressing well BID-QID • 5 minutes of uphill inclines—<i>introduce at 3-4 weeks postoperatively</i> • Begin slow jogging 2-5 minutes daily (<i>begin at weeks 3-4 postoperatively</i>) • Aquatic therapy—Underwater Treadmill (UWTM) walking, begin at 5-10 min up to 15-30 min. (Ensure <i>incision is sealed before beginning aquatic therapy</i>) • Balance exercises on a 360-degree wobble board or moderately unstable surface 5-10 min	Weight loss program can be initiated at this phase if warranted (minimizes loss of lean tissue mass). Therapeutic Exercises • PROM and flexion/extension stifle stretches of operated limb (10-20 reps BID-TID) if normal ROM not yet achieved • Slow consistent leash walking—add 5 min every week if progressing well BID-QID • 5 minutes of uphill inclines <i>introduce at 3-4 weeks postoperatively</i>	• Response to activity and subjective pain level • Gait scale at walk and trot • Goniometry—Stifle ROM and other joints if applicable • Reassess thigh muscle girth at 2, 4, and 6 weeks postoperatively • Weight • Consistent weight-bearing on operated limb at a trot • No evidence of pain with activities • ROM 40-60 degrees flexion, >150 degrees extension

CCL Rupture—Extracapsular Repair—cont'd

Phase	Expected Timeframe*	Rehabilitation Clinic Program	Home Program	Outcome Assessment Measures	Criteria for Movement to Next Phase
		• Sit-to-stand exercises—5-10 reps working up to 10-15 reps BID—<i>begin at 4 weeks postoperatively</i> • Stair climbing—1-2 reps up and down working up to 5-10 reps BID—<i>begin at 4-6 weeks postoperatively</i> • Aquatic therapy—Underwater Treadmill (UWTM) walking, begin at 5-10 min up to 15-30 min • Swimming 2-5 minutes two-three times per week—<i>begin at 4-6 weeks postoperatively</i> • Cavaletti Rails 5 reps × 6 rails to begin <i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i> Modalities • Ultrasound therapy if loss of tissue extensibility—perform in conjunction with PROM and stretching if needed • Therapeutic laser PRN • Cryotherapy (15-20 min BID) following exercises if needed	• Begin slow jogging 2-5 minutes daily (<i>begin at weeks 3-4 postoperatively</i>) • Sit-to-stand exercises—5 reps worked up to 20 reps BID—<i>begin at 4 weeks postoperatively</i> • Stair climbing—1-2 reps up and down working up to 5—<i>begin at 4-6 weeks postoperatively</i> Medications • NSAIDs—PRN • Injectable polysulfated glycosaminoglycans • Chondroprotective agents		
Step 4 Consistent Weight-Bearing at a Trot <u>Goals:</u> Improve ability to use the limb with some speed	Six to twelve weeks postoperatively	Therapeutic Exercises • Trotting 5-10 min BID, increasing to 15-25 min TID, as long as pain and lameness do not worsen • Sit-to-stand exercises—15-20 reps BID • Aquatic therapy—brisk UWTM walking • Swimming several times per week • Cavaletti rails 15-20 reps BID, gradually raising rails	Continue weight loss program if needed Therapeutic Exercises • Trotting program, beginning at 5-10 min BID working up to 15-25 min TID • Uphill inclines jogging, beginning at 1-2 minutes and working up to 5-10 minutes per session	• Response to activity and subjective pain level • Gait evaluation at walk and trot	• Minimal to no lameness at a trot of normal speed • No evidence of pain with activities

Improve muscle mass and strength Improve stamina and endurance	• Ground treadmill walking with leg weights and/or resistance bands <i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i> Modalities • Heat therapy (10-15 min BID-TID, if needed for stretching) • Cryotherapy (15-20 min BID) following exercises if needed	• Controlled ball play at fast trot or slow lope (alternate days with trotting program) • Sit-to-stand exercises, 15-20 reps BID • Stair climbing—1-2 flights 2-5 reps as tolerated <i>Caution: Choose only 2-4 exercises per session. Doing all exercises may result in lameness and pain early in the program</i> Medications • NSAIDs—PRN • Injectable polysulfated glycosaminoglycans • Chondroprotective agents	• Reassess thigh muscle girth at 10-12 weeks postoperative • Weight limits (depending on pre-operative values)
<u>Step 5</u> Trotting at Speed with Minimal to No Lameness <u>Goals:</u> Continue improving speed at a lope and slow gallop Continue improving muscle mass, strength and power Continue improving stamina and endurance Return to as normal activity as possible	Twelve weeks postoperatively and beyond Therapeutic Exercises • Trotting/running 10-20 min BID, as long as pain and lameness do not worsen. Some running up hills • Trotting up stairs 1-2 flights, 2 repetitions BID • Ground treadmill jogging with leg weights and/or resistance bands • Aquatic therapy—UWTM for conditioning, consider progressively increasing speed and use of jets when available • Swimming 10-15 minutes 3-4 times per week <i>Caution: Choose only 2-3 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i> Modalities • Heat therapy (10-15 min BID-TID, NOT within 72 hours after surgery) • Therapeutic laser PRN • Cryotherapy (15-20 min BID) following exercises if needed	Therapeutic Exercises • Running up and down hills • Ball playing at a gallop, 3-5 min working up to 15-20 minutes daily • Jogging for prolonged distances regularly throughout the week • Trotting up stairs—1-2 flights 2-3 reps as tolerated <i>Caution: Choose only 1-2 exercises per session. Doing all exercises may result in lameness and pain early in the program</i> Medications • Polysulfated glycosaminoglycans • NSAIDs—PRN • Chondroprotective agents	Progressions may be continued as long as lameness or pain with activity is not observed. If this occurs, decrease activity by 50% for 1 week and increase again by 10-20% weekly.

This protocol is for use only by licensed personnel who have been trained in canine rehabilitation.

*The therapist must keep in mind that these timelines are intended as guidelines and may be influenced by many factors affecting individualized patient care.

CCL Rupture—TPLO/TTA

Phase	Expected Timeframe*	Rehabilitation Clinic Program	Home Program	Outcome Assessment Measures	Criteria for Movement to Next Phase
Preoperative	From diagnosis to time of surgery	Aquatic Therapy Exercises - Brief training to aquatic therapy to ensure patient comfort and compliance Medications - Injectable polysulfated glycosaminoglycan agents (<i>discontinue within 24 hours of surgery</i>) Modalities - Transcutaneous electrical stimulation for pain relief (15-20 min SID) - NMES for muscle strengthening (15 min SID PRN) - Cryotherapy (15-20 min BID)	Therapeutic Exercises - Rest in a confined space - Slow, controlled leash-walking only to go outside, no more than five minutes at a time Medications - NSAIDs (<i>discontinue within 48 hours of surgery</i>) - Oral chondroprotective agents Modalities - Cryotherapy (15-20 min BID)	- Baseline bilateral stifle ROM—goniometry; other joints if applicable - Baseline bilateral thigh circumference - Weight	Surgery should be performed as soon as possible after diagnosis to reduce the onset and severity of osteoarthritis as a sequela of motion on an unstable stifle.
Step 1 Non-Weight-bearing to Toe-Touching	Immediate to 48-72 hours postoperative	Therapeutic Exercises - PROM for stifle flexion/extension of operated limb (10 reps BID-TID or more if PROM is poor)** - Slow leash walking with sling support available, only to go outside (up to 5 minutes, 2-3 times per day) Medications - NSAIDs (<i>administration 30-60 minutes prior to first therapy session recommended</i>) - Opioids PRN Modalities - Mild pressure bandage (<i>first 24 hours postoperative</i>) - Gentle massage and thigh/stifle soft tissue mobilization - Transcutaneous electrical stimulation for pain relief (15-20 min SID-TID) - Cryotherapy (15-20 min BID)—<i>first session immediately postoperative</i> - Therapeutic laser therapy SID	Inpatient status during this phase - Postoperative bilateral stifle ROM—goniometry; other joints if applicable - Postoperative bilateral thigh circumference - Response to activity and subjective pain level - Lameness score at a stance and walk - Weight	- Early toe-touching - Adequate resting analgesia - Decreased perioperative swelling and lack of incisional drainage - ROM 60-80 degrees flexion, 120-135 degrees extension	

Step 2

Early Weight-Bearing

72 hours to three weeks postoperative

Therapeutic Exercises

- PROM and flexion/extension stifle stretches of operated limb (10-15 reps BID-TID) until near normal stifle ROM achieved
 - Slow, controlled leash walking, 5 minutes working slowly up to 20 minutes at a time TID
 - Balance exercises on a soft foam pad or bi-directional balance board for weight-bearing
 - Aquatic therapy—Underwater Treadmill (UWTM) walking 5-10 min SID-BID (*do not begin until incision is sealed, approximately 1-2 weeks postoperative*)
- Modalities
- Heat therapy before activity (10-15 min BID-TID, NOT within 72 hours after surgery or if CS of acute inflammation are still present)
 - Therapeutic laser therapy PRN
 - Cryotherapy (15-20 min BID) following exercises

Therapeutic Exercises

- PROM and flexion/extension stifle stretches of operated limb (10-15 reps)
 - Slow, controlled leash walking, 5 minutes working slowly up to 20 minutes at a time TID
- Medications
- NSAIDs—PRN
 - Wean off Tramadol, NMDAs
 - Oral chondroprotective agents
- Modalities
- Heat therapy (10-15 min BID-TID, prior to exercises, NOT within 72 hours after surgery or if CS of acute inflammation are still present)
 - Cryotherapy (15-20 min BID) following exercises

Goniometry—

- Stifle ROM and other joints if applicable
- Reassess thigh muscle girth at 3 weeks
- Postoperative Response to activity and subjective pain level
- Lameness score at a stance and walk
- Weight

Consistent

- Goniometry—partial weight-bearing on operated limb on all strides at a walk
- Minimal pain with light activities
- Incisional healing without complications
- ROM 50-60 degrees flexion, >150 degrees extension

Step 3

Consistent Weight-Bearing

Three to six weeks postoperative

Therapeutic Exercises

- PROM and flexion/extension stifle stretches of operated limb (10-20 reps BID-TID) if normal ROM not yet achieved
- Balance exercises on a 360-degree wobble board or moderately unstable surface 5-10 min
- Sit-to-stand exercises—5 working up to 10-15 reps BID-TID—operated side against a wall
- Aquatic therapy—Underwater Treadmill (UWTM) walking working from 5-10 min up to 15-30 min SID-BID
- Low Cavaletti Rails 5 reps × 6 rails to begin

Weight loss program can be initiated at this phase if warranted (minimizes loss of lean tissue mass).

Therapeutic Exercises

- Slow consistent leash walking—working up to 20-30 minutes BID-TID if progressing well
- 5-10 minutes of uphill inclines introduced at 4-5 weeks postoperative
- Sit-to-stand exercises—5 working up to 10-15 reps BID-TID—operated side against a wall

Response to

- activity and subjective pain level
- Gait scale at walk
- Goniometry—limb at a walk
- No evidence of pain with activities
- ROM 40-60 degrees flexion, >150 degrees extension

Continued

CCL Rupture—TPLO/TTA—cont'd				
Phase	Expected Timeframe*	Rehabilitation Clinic Program	Home Program	Outcome Assessment Measures
		Modalities • Ultrasound therapy if loss of tissue extensibility—perform in conjunction with PROM and stretching if needed • Therapeutic laser therapy PRN • Cryotherapy (15-20 min BID) following exercises if needed	• Walking in circles to left and right or Figure 8s to improve proprioception (<i>beginning with wider and working towards narrower turns</i>) Medications • NSAIDs—PRN • Chondroprotective agents	Criteria for Movement to Next Phase
Step 4 Improved Weight-Bearing at a Trot	Six to eight weeks postoperative	Therapeutic Exercises • <i>Radiographs to demonstrate healing of osteotomy site are required to proceed.</i> • Sit-to-stand exercises—20-30 reps BID • Aquatic therapy—UWTM walking • Swimming several times per week, 5 up to 20 min per session (<i>gentle water entry/exit required</i>) • Cavaletti rails 10-20 reps BID, gradually reducing distance between rails • Stair climbing: 5-10 stairs up and down, BID-TID • Controlled tug-of-war, 5 min per session to tolerance SID Modalities • Heat therapy (10-15 min BID-TID) prior to exercises if needed • Cryotherapy (15-20 min BID) following exercises if needed	Therapeutic Exercises • <i>Radiographs to demonstrate healing of osteotomy site are required to proceed.</i> • Prolonged leash walks, 30-40 min BID-TID, longer leash allowed • Controlled on-leash trotting, beginning at 5 working up to 10 min BID, in a straight line • Trotting in Figure 8s, no sharp turns • Sit-to-stand exercises, 20-30 reps BID • Stair climbing—5-10 stairs up and down, BID-TID • Dancing on rear limbs 30 sec-1 min per rep, BID-TID • Controlled tug-of-war, 5 min per session to tolerance SID Medications • NSAIDs—PRN • Chondroprotective agents	• Response to activity and subjective pain level • Gait scale at walk and trot • Reassess thigh muscle girth at 7-8 weeks postoperative • Weight • Consistent weight-bearing at a trot • No evidence of pain with activities

Step 5 Consistent Weight- Bearing at a Trot	Nine weeks post- operative and beyond	Therapeutic Exercises • Aquatic therapy—UWTM for conditioning, consider progressively increasing speed and use of jets when available • Swimming 15-20 minutes 3-4 times per week • Weights on affected limb (begin at 0.5-1 lb depending on dog size) or therabands for preferential strengthening • Playing ball • Jogging up hills and stairs • Additional therapeutic exercises as indicated for strength and speed Modalities • Cryotherapy (15-20 min BID) following exercises if needed	Therapeutic Exercises • Program may continue as before with the below additions • Running in a straight line • Trotting in Figure 8s and zigzag patterns • Weights on affected limb (begin at 0.5-1 lb depending on dog size) for preferential strengthening • Stair climbing—1-2 flights 1-2 reps as tolerated Medications • NSAIDs—PRN • Chondroprotective agents	• Response to activity and subjective pain level • Gait scale at walk and trot, eventually at faster gaits • Reassessment of thigh girth and joint goniometry as applicable • Weight Goals are based on multifactorial considerations, including age, presence/degree of osteoarthritis, and musculoskeletal status of other limbs.	Progressions may be continued as long as lameness or pain with activity is not observed. If this occurs, decrease activity by 50% for 1 week and increase again by 10-20% weekly.
---	--	--	--	---	--

*This protocol is for use by licensed personnel who have been trained in canine rehabilitation. The therapist must keep in mind that these timelines are intended as guidelines and may be influenced by many factors affecting individualized patient care.

**Loss of extension range of motion is more common than flexion after a tibial plateau leveling osteotomy.

Cervical Spondylomyelopathy (postoperative)

Phase	Expected Timeframe*	Rehabilitation Clinic Program	Home Program	Outcome Assessment Measures	Criteria for Movement to Next Phase
Rest Goals: Pain control Maintenance of joint ROM Maintain incisional health Weight-bearing in standing with moderate support Minimize limb muscle atrophy Independent urination and defecation	Immediate to five to seven days postoperatively	Therapeutic Exercises - Assisted standing with physioball abdominal support on a stable, high-traction surface (<i>beginning 48-72 hours postoperatively</i>) - PROM with bicycling movements all joints of all four limbs (10-15 reps BID-TID). - Flexor reflex stimulation exercises (8-10 reps BID-TID for each limb) Modalities - TENS, premodulated or interferential electrical stimulation across incision site, 15-20 min SID - NMES to scapular and/or thigh muscles 15 min SID - Cryotherapy at cervical incision site, 20 min BID-TID (<i>first 48-72 hours after surgery</i>) - Heat therapy at cervical incision site, 10 min BID-TID (<i>after 72 hours postoperatively</i>) - Gentle massage of shoulder and limb musculature +/- tapotement BID as tolerated - Therapeutic laser therapy of peri-surgical cervical and shoulder musculature SID PRN	Therapeutic Exercises - Assisted standing with thoracic and abdominal support as needed on a stable, high-traction surface (<i>beginning 48-72 hours postoperatively</i>) - PROM with bicycling movements all joints of all four limbs (10-15 reps BID-TID) Modalities - Cryotherapy at cervical incision site, 20 min BID-TID (<i>first 48-72 hours after surgery</i>) - Heat therapy at cervical incision site, 10 min BID-TID (<i>after 72 hours postoperatively</i>) - Effleurage BID-TID Medications - NSAIDs (if corticosteroids have not been used)/opioids - NMDA receptor antagonists - Muscle relaxants BID-TID PRN Other - Bladder expression TID-QID PRN - Use of soft bedding with rotation every 4-6 hours to prevent decubital ulcers	- Subjective pain level - Limb muscle tone - Degree of support required in standing - Limb muscle circumference (baseline) - Subjective scapular muscle mass	- Postoperative comfort - Incisional integrity maintained - Standing in weight-bearing with moderate support

Other		Note: A home exercise program may be impractical for many owners early in the postoperative period, depending on patient size and function		
Partial Weight Support to Initial Motor Function Goals: Continued pain control Weight-bearing in standing with minor support Minimize muscle atrophy Voluntary movement of all four limbs, with supported early walking Self-righting into a sternal position	One to two weeks postoperatively	Therapeutic Exercises - Assisted standing and weight-shifting/rhythmic stabilization, gradually decreasing support (eg physioball beneath abdomen, standing in water) - PROM with bicycling movements all joints of all limbs (10-15 reps BID-TID)—during supported standing if feasible - Early walking with sling support (patient capability will vary) short distances, eg 1-2 yoga mat lengths up to 2-3 minutes Modalities - Gentle massage of limb musculature - BID-TID - TENS, premodulated or interferential electrical stimulation across incision site, 15-20 min PRN - NMES to hindlimb and forelimb musculature as needed for contraction 15 min EOD-SID - Heat therapy at cervical incision site, 10 min BID-TID (after 72 hours postoperatively) Other - Bladder expression TID-QID PRN - Use of soft bedding with rotation every 4-6 hours to prevent decubital ulcers	Therapeutic Exercises - Assisted standing and weight-shifting/rhythmic stabilization, gradually decreasing support (eg physioball beneath abdomen) - PROM with bicycling movements all joints of all limbs (10-15 reps BID-TID) - Flexor reflex stimulation (8-10 reps BID-TID for each limb) Modalities - Gentle massage of hindlimb musculature - BID-TID - Heat therapy at cervical incision site, 10 min BID-TID - Medications - NSAIDs (if corticosteroids have not been used)/opioids - NMDA receptor antagonists Other - Bladder expression TID-QID PRN - Use of soft bedding with rotation every 4-6 hours to prevent decubital ulcers	- Response to activity and subjective pain level - Degree of support required in standing - Supported walking endurance (seconds to minutes) - Degree of appropriate gait patterning in motion - Limb muscle circumference/subjective scapular muscle mass - Degree of comfortable active cervical ROM - Voluntary motor activity in all four limbs - Early stepping activity all four limbs (supported walking) for short distances - Incisional healing - Minimal pain with light activities - Standing with minor support

Cervical Spondylomyelopathy (postoperative)—cont'd

Phase	Expected Timeframe*	Rehabilitation Clinic Program	Home Program	Outcome Assessment Measures	Criteria for Movement to Next Phase
Initial Motor Function to Good Motor Function with Proprioceptive Deficits <u>Goals:</u> Improve proprioception during walking activity Maintain normal limb ROM Increase limb muscle mass Maintenance of balance during turns or on a moderately unstable surface when standing	Two to four weeks postoperatively	Therapeutic Exercises - Supported standing and weight-shifting on a bi-directional balance board - Standing PROM with bicycling movements all joints of all limbs (10-15 reps PRN based on ROM) - Aquatic therapy: Swimming 5-10 minutes with supported standing during frequent breaks SID. <i>Cervical patients should wear lifejacket and be accompanied by assisting personnel in the pool at early stages of use</i> - Aquatic therapy: Underwater Treadmill (UWTM) standing (up to 10 minutes) to slow walking (up to 5 minutes with breaks) SID. <i>Cervical patients should wear lifejacket and be accompanied by assisting personnel in the UWTM at early stages of use</i> - Turning exercises during standing with positive reinforcement held at various locations (BID-TID, concentrate efforts on the weaker side)—caution with cervical ROM - Sit-to-stand exercises while sling-supported with hindquarters sitting on an elevated surface <i>Caution: Choose only 2-3 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i> Modalities - NMES to hindlimb musculature for contraction 15 min PRN - Cryotherapy at cervical incision site, 20 min PRN after exercise	Therapeutic Exercises - Supported standing and weight-shifting on a moderately unstable surface - Standing PROM with bicycling movements all joints of all limbs (10-15 reps PRN based on ROM) - Slow leash walking with sling and forelimb harness support 5-10 minutes BID-TID - Turning exercises during standing with positive reinforcement held at various locations (concentrate efforts on the weaker side) Modalities - Cryotherapy at cervical incision site, 15-20 min PRN after exercise - Medications - NSAIDs PRN - NMDA receptor antagonists	- Response to activity and subjective pain level - Length of time with minimally supported/unsupported standing - Length of time walking in aquatic therapy and on dry land - Frequency of steps in which appropriate proprioception is used - Limb muscle circumference/subjective scapular muscle mass - Degree of comfortable active cervical ROM	- Consistent gait patterning of all limbs at a slow walk - Minimal pain with moderate activities - Walking endurance of 5-10 minutes with breaks - Early appropriate proprioceptive limb placement

Good motor function with proprioceptive deficits to near normal gait <u>Goals:</u> Consistent proprioception with slow-moderate activity Limb muscle mass approaching normal Comfortable cervical AROM approaching normal Maintenance of balance during turns and negotiating obstacles while walking	Four to eight weeks postoperatively	Therapeutic Exercises - Assisted standing and weight-shifting on a 360-degree balance board - Aquatic therapy: Swimming 10-15 minutes with supported standing during frequent breaks - Aquatic therapy: Underwater Treadmill (UWTM) walking, 5-10 minutes SID with lifejacket or harness - Gait patterning on a ground treadmill, 5-10 minutes with resistance band use - PRN to encourage proprioceptive placing - Turning exercises when walking: weave poles gradually reduced in distance 5-10 reps SID-BID - Low cavaletti rails 5-10 reps SID-BID - Sit-to-stand exercises, rising from an elevated surface 10-15 reps SID-BID <i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in lameness and pain early in the program</i>	Therapeutic Exercises - Assisted standing and weight-shifting on an unstable surface - Walking in circles and figure-eights 3-5 minutes BID-TID - Sit-to-stand exercises, rising from an elevated surface 5-10 reps SID-BID - Sling-supported leash walking 10-15 min BID-TID with 3-5 minutes gentle inclines/declines - Turning/stretching exercises during standing with positive reinforcement held at various locations (concentrate efforts on the weaker side) Medications - NSAIDs PRN - NMDA receptor antagonists	- Response to activity and subjective pain level - Length of time with minimally supported/unsupported standing - Walking endurance - Frequency of steps in which appropriate proprioception is used - Limb muscle circumference/subjective scapular muscle mass - Height of Cavaletti rails navigated - Tightness of turns successfully executed - Cervical AROM	- Consistent pelvic limb proprioception at a walk - Minimal to no pain with moderate controlled activities - Normal gait patterning at a walk - Near-normal active cervical ROM
--	-------------------------------------	--	---	--	--

Continued

Cervical Spondylomyelopathy (postoperative)—cont'd

Phase	Expected Timeframe*	Rehabilitation Clinic Program	Home Program	Outcome Assessment Measures	Criteria for Movement to Next Phase
Near Normal Gait to Normal Gait at a Walk <u>Goals:</u> Elimination of proprioceptive deficits during controlled walking and trotting Normal limb muscle mass Normal cervical AROM/PROM Maintenance of balance during turns and negotiating obstacles while trotting	Eight or more weeks postoperatively	Therapeutic Exercises • Standing and weight-shifting on a 360-degree balance board • Aquatic therapy: Swimming 15-20 minutes SID • Aquatic therapy: Underwater Treadmill (UWTM) walking, 15-20 minutes working up to variable speeds • Gait patterning on a ground treadmill, 5-10 minutes walking working up to a trot • Turning exercises with weave poles 10-20 reps SID-BID at a walk working up to a trot • Higher Cavaletti rails 10-20 reps SID-BID • Stair Climbing beginning with 1-3 flight repetitions SID-BID • Sit-to-stand exercises 10-20 reps SID-BID <i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Gradual increase to normal activity can be initiated as early as 8 weeks postoperatively, but many patients may take significantly longer to reach the goals listed at this stage.</i>	Therapeutic Exercises • Standing and weight-shifting on an unstable surface • Turning exercises with weave poles 10-20 reps SID-BID at a walk working up to a trot • Stair Climbing 1-3 flight repetitions SID-BID • Sit-to-stand exercises 10-20 reps SID-BID • Controlled ball play in a confined space • Leash walking 15-20 minutes BID with 5-8 minutes mild to moderate inclines and 3-5 minutes controlled trotting (working up to longer periods)	• Response to activity and subjective pain level • Walking and trotting endurance • Tightness of turns in which balance is maintained at a walk and trot • Height and distance of Cavaletti rails navigated • Limb muscle circumference/muscle mass • Proprioceptive placing at a controlled trot	• Minimal to no pain with moderate-to-extensive controlled activities • Normal gait and proprioception at a controlled trot • Normal comfortable cervical AROM/PROM

This protocol is for use by licensed personnel who have been trained in canine rehabilitation.

*The therapist must keep in mind that these timelines are intended as guidelines. Rate of recovery will vary widely among patients for reasons that include status of nociception prior to surgery.

Degenerative Myelopathy^{*,†}

Phase	Expected Timeframe*	Rehabilitation Clinic Program	Home Program	Outcome Assessment Measures	Criteria for Movement to Next Phase
Early Stage: Pelvic limb paraparesis and ataxia Goals: Minimize loss of muscle strength/endurance Maintain quality of life Prolong survival time	Variable	Therapeutic Exercises - Assisted standing and weight-shifting on a 360-degree balance board 5-10 min BID - Aquatic therapy: Underwater Treadmill (UWTM) walking, 5-10 working up to 20-30 min at variable speeds SID-BID - Swimming 10 working up to 30 minutes with breaks - Sit-to-stand exercises 10-15 reps BID - Cavaletti Rails (walking or trotting as able) 10-15 reps BID - Weave Poles (walking or trotting as able) 10-15 reps BID - Walking up and down stairs 2-4 flights BID-TID - Land treadmill walking/jogging as capable 5-10 min SID-BID. Add incline to challenge pelvic limbs or decline to facilitate appropriate movement of pelvic limbs <i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in over-exertion and pain early in the program</i> Modalities - Gentle massage of limb and paraspinal musculature SID-BID prior to or following exercise - NMES Quadriceps and Hamstrings 15 min SID-EOD Other - Consider use of paw protective boots as proprioceptive deficits develop	Therapeutic Exercises - Hemi-standing or diagonal leg two-legged standing with weight-shifting 5-10 min divided into 1-2 min reps BID-TID - Leash walks at routinely performed pace/distance, incorporating inclines and 5 minutes of trotting as able BID-TID - Controlled ball play - Sit-to-stand exercises 10-15 reps BID - Walking/trotting as capable across an unstable surface such as a foam mattress 10-15 laps SID-BID - Walking up and down stairs 4-6 flights BID-TID - Walking/trotting as capable through sand or tall grass 5 min SID-BID <i>Caution: Choose only 2-4 exercises per session. Doing all exercises may result in over-exertion and pain early in the program</i> Modalities - Effleurage BID-TID Other - Consider use of paw protective boots as proprioceptive deficits develop	- Response to activity and subjective pain/ exertion level - Walking and trotting endurance on dry land - Walking endurance in underwater treadmill - Frequency of steps in which appropriate proprioception is used (if deficits are present) - Height of Cavaletti rails navigated - Thigh circumference/ muscle mass	- Inconsistent pelvic limb proprioception at a walk - Moderate paraparesis/ ataxia at a walk - Severe paraparesis/ ataxia at a trot - Moderate-severe paraparesis/ ataxia when turning

Continued

Degenerative Myelopathy**—cont'd

Phase	Expected Timeframe*	Rehabilitation Clinic Program	Home Program	Outcome Assessment Measures	Criteria for Movement to Next Phase
Early-Mid-Stage: Moderate to severe pelvic limb paraparesis and ataxia <u>Goals:</u> Delay loss of muscle strength/endurance Provide continued proprioceptive feedback Maintain quality of life Prolong survival time	Variable	Therapeutic Exercises • PROM with bicycling movements all joints of pelvic limbs (10-15 reps BID-TID) • Flexor reflex exercises for pelvic limbs, 10 reps BID • Assisted standing and weight-shifting on a foam pad or bi-directional balance board with assisted pelvic limb support as needed • Aquatic therapy: UWTM walking, 5-15 min at a steady pace SID-BID • Swimming 10-15 minutes with breaks 3-4 times per week • Sit-to-stand exercises with sling assistance if needed 10-15 reps BID • Low Cavaletti Rails 5-10 laps as able BID • Weave Poles (walking as able) 10-15 reps BID • Land treadmill slow walking as capable 5-10 min SID-BID. Add decline to reduce weight-bearing forces on pelvic limbs, +/- resistance band to help encourage proprioceptive placing <i>Caution: Choose only 2-3 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in over-exertion and pain early in the program</i>	<i>Fitting for a four-wheel cart if desired can take place at this time</i> Therapeutic Exercises • PROM with bicycling movements all joints of pelvic limbs (10-15 reps BID-TID) • Flexor reflex exercises for pelvic limbs, 10 reps BID • Balance and weight-shifting on a moderately unstable surface with pelvic limb support as needed; 5-10 min divided into 1-2 min reps BID-TID • Walks (with four-wheel cart when prescribed) at routinely performed pace/distance on level ground, 10-15 min BID-TID • Walks in front and rear assistive harness with resistance bands at the tarsus/metatarsal bones pulling craniodorsally • Sit-to-stand exercises with sling assistance if needed 10-15 reps BID • Assisted walking through sand or tall grass 5 min SID-BID <i>Caution: Choose only 2-3 exercises per session. Doing all exercises may result in over-exertion and pain early in the program</i> Modalities • Effleurage BID-TID	• Response to activity and subjective pain/exertion level • Walking endurance and frequency of proprioceptive placing in underwater treadmill • Presence or absence of proprioception at a walk • Ability to support weight in standing • Height of obstacles navigated • Pelvic limb voluntary movement • Thigh circumference/muscle mass	• Poor pelvic limb proprioception at a walk • Severe paraparesis or paraplegia at a walk • Extensive assistance required for pelvic limbs in weight-bearing

Modalities - Gentle massage of limb and paraspinal musculature SID-BID prior to or following exercise - NMES Quadriceps and Hamstrings 15 min SID-EOD Other Use paw protective boots as proprioceptive deficits increase		Other Use paw protective boots as proprioceptive deficits increase	
Late-Stage: Pelvic limb paraplegia Goals: Reduce loss of muscle strength/endurance Provide continued proprioceptive feedback Maintain quality of life Prolong survival time	Variable; 6-9 months post-diagnosis	Therapeutic Exercises - PROM with bicycling movements all joints of all limbs (10-15 reps BID-TID) - Flexor reflex exercises for pelvic and thoracic limbs, 10 reps BID - Aquatic therapy: Swimming 10-15 minutes with supported standing during breaks - Aquatic therapy: UWTM walking, 10-15 minutes SID. Use flotation devices/resistance bands to assist with proprioceptive placing if needed. Modalities - NMES to hindlimb musculature for contraction 15 min EOD-SID PRN	Therapeutic Exercises - PROM with bicycling movements all joints of all limbs (10-15 reps BID-TID) - Flexor reflex exercises for pelvic and thoracic limbs, 10 reps BID - Walks (with four-wheel cart) at routinely performed pace/distance on level ground, 10-15 min BID-TID. <i>Use protective boots on pelvic limbs in cart unless limbs are completely suspended</i> Proprioception and postural reactions of thoracic limbs - Pelvic/thoracic limb reflexes - Subjective thoracic limb muscle mass - Patient attitude Complete loss of voluntary motion of pelvic limbs - Early, mild thoracic limb paraparesis
Late Stage: Pelvic Limb Paraplegia with Thoracic Limb Paraparesis Goals: Maintain patient comfort	Variable; up to 19-20 months post-diagnosis	Therapeutic Exercises - PROM with bicycling movements all joints of all limbs (10-15 reps BID-TID) - Flexor reflex exercises for pelvic and thoracic limbs, 10 reps BID - Aquatic therapy: Swimming 10-15 minutes with supported standing during breaks. <i>Personnel assistance should be provided in the pool to achieve limb movement and for safety</i> Other - Bladder expression TID-QID PRN - Use of soft bedding with rotation every 4-6 hours to prevent decubital ulcers	Therapeutic Exercises - PROM with bicycling movements all joints of all limbs (10-15 reps BID-TID) - Flexor reflex exercises for pelvic and thoracic limbs, 10 reps BID - Bladder expression TID-QID PRN - Use of soft bedding with rotation every 4-6 hours to prevent decubital ulcers Pelvic/thoracic limb reflexes - Patient attitude End-stage of the disease with variable urinary and/or fecal incontinence and probability of eventual dysphagia or respiratory arrest

This protocol is for use by licensed personnel who have been trained in canine rehabilitation.

*The therapist must keep in mind that these timelines are intended as guidelines. Rate of progression will vary widely among patients for reasons that include initial status of nociception and degree of neurologic deficits at the time of injury.

**Exercise regimen proposed to prolong survival time in dogs affected by degenerative myelopathy is based on Kathmann, I. et al. Daily Controlled Physiotherapy Increases Survival Time in Dogs with Suspected Degenerative Myelopathy. *J Vet Intern Med* 2006; 20: 927-932.

Distal Femoral Physeal Fractures

Phase	Expected Timeframe*	Rehabilitation Clinic Program	Home Program	Outcome Assessment Measures	Criteria for Movement to Next Phase
Immobilization <u>Goals:</u> Pain control Swelling and edema control Encourage/improve ROM and hindlimb extension	Immediate to 48-96 hours postoperatively	Therapeutic Exercises • PROM and gentle stifle flexion/extension (10-15 reps TID-QID). <i>Specific attention should be paid to stifle flexion</i> • Elicit flexor reflex with interdigital manipulation 10 reps BID-TID • Balance and weight-shifting exercises BID-TID to encourage weight-bearing and active stifle flexion/extension on the operated limb Modalities • Cryotherapy (15-20 min (<i>first session immediately postoperative</i>) TID) • TENS across fracture site SID-BID. Consider edema electrical stimulation protocol Medications • NSAIDs—daily PRN • Opioids—BID-TID PRN	Therapeutic Exercises • PROM and gentle stifle flexion/extension (10-15 reps TID-QID). <i>Specific attention should be paid to stifle flexion</i> • Balance/weight-shifting exercises to encourage toe-touching on a stable surface Immobilization between sessions for 2-4 days in a 90-degree/90-degree stifle/tarsal flexion sling** Modalities • Cryotherapy (15-20 min BID) following exercises	• Weight • Goniometry—stifle ROM • Swelling/edema • Response to PROM and subjective pain level • Early toe-touching to partial weight-bearing	• Postoperative comfort • Incisional integrity maintained • Minimal pain with PROM • Early toe-touching to partial weight-bearing
Early Weight-Bearing <u>Goals:</u> Continued pain control Near-normal ROM and hindlimb extension Minimize muscle atrophy of operated limb Achieve consistent partial weight-bearing on operated limb	48-96 hours to 10 days postoperatively	Therapeutic Exercises • PROM bicycling movements with mild-moderate resistance and stifle flexion/extension stretching (10-15 reps BID-TID). <i>Specific attention should be paid to stifle flexion</i> • Slow, controlled leash or land treadmill walking, 5-10 min BID-TID to encourage active range of motion in the stifle • Balance and weight-shifting exercises BID-TID to encourage weight-bearing and active stifle flexion/extension on the operated limb	Therapeutic Exercises • PROM and flexion/extension hip stretches of operated limb (10-15 reps BID-TID) • Slow, controlled leash walking, 5-10 min TID-QID to encourage active range of motion in the stifle • Balance exercises on a soft foam pad for weight-bearing BID-TID Medications • NSAIDs—daily PRN • Opioids—BID-TID PRN	• Goniometry—stifle ROM and other joints if applicable • Thigh muscle girth/symmetry • Response to activity and subjective pain level • Lameness score at a stance and walk • Weight	• Consistent partial weight-bearing on operated limb on all strides at a walk • Minimal pain with light activities • Incisional healing without complications

• Use of a syringe cap or other non-harmful irritant on contralateral foot during exercise to improve operated limb weight-bearing • Continued sling use to discourage stifle hyperextension; if present, should be considered with removal only for rehabilitation sessions Modalities • Heat therapy to quadriceps muscles 10 min prior to PROM/exercise BID-TID • Massage of quadriceps muscles before active exercise • TENS across fracture repair site for analgesia SID-BID • Therapeutic laser PRN • Cryotherapy (15-20 min PRN) following exercises	Consistent Weight-Bearing at a Walk Goals: - Minimize pain during low- to moderate-intensity activity - Maintain normal ROM and hindlimb flexion and extension - Reduce and prevent further muscle atrophy - Achieve minimal lameness at a walk Ten days to three weeks postoperatively Therapeutic Exercises • PROM and gentle stifle flexion/extension (10-15 reps BID-TID), continued through 3-4 weeks postoperative • Underwater treadmill walking, 10-15 min SID-BID to encourage active range of motion in the stifle • Swimming 8-10 minutes SID-BID if hindlimbs are used • Balance and weight-shifting exercises BID-TID on increasingly unstable surfaces to improve muscular adjustments and active stifle flexion/extension on the operated limb Therapeutic Exercises • PROM and flexion/extension hip stretches of operated limb (10-15 reps BID-TID) • Slow, controlled leash walking, 10-15 min TID-QID to encourage active range of motion in the stifle • Include inclined surface into slow leash walks (3-5 minutes with each walking session) for quadriceps and gluteal muscle strengthening • Balance exercises on an unstable surface for weight-bearing BID-TID Goniometry—stifle ROM and other joints if applicable • Response to activity and subjective pain level Consistent substantial weight-bearing on operated limb on all strides at a walk • Minimal pain with moderate activities • Near-normal stifle passive ROM • Minimal to no stifle effusion
--	--

Continued

Distal Femoral Physal Fractures—cont'd

Phase	Expected Timeframe*	Rehabilitation Clinic Program	Home Program	Outcome Assessment Measures	Criteria for Movement to Next Phase
Partial Weight-Bearing at a Trot Goals: Pain-free moderate- to high-intensity activity Normal muscle mass of operated hindlimb Achieve soundness at a trot	Three to eight weeks postoperatively	Modalities • Heat therapy to affected quadriceps BID-TID, 10 minutes • Massage of quadriceps muscles before active exercise PRN • Cryotherapy (15-20 min PRN) following exercises	Medications • NSAIDs—PRN • Opioids—SID-BID PRN Modalities • Heat therapy PRN before activity • Cryotherapy (15-20 min BID) following exercises	• Lameness score at a stance and walk • Weight	• Radiographic evidence of bone healing 3-4 weeks postoperatively
		Therapeutic Exercises • Aquatic therapy: UWTM walking 15-20 minutes SID-BID • Swimming 20-30 minutes SID-BID • Balance exercises on an inflatable disk or 360 degree wobble board for weight-bearing 10 min BID-TID • Land treadmill walking with caudal pull from resistance band to encourage quadriceps strengthening: 10-15 min SID-BID, wait until 4-6 weeks postoperative <i>Note: Ensure radiographic evidence of fracture healing and implant integrity preceding to these exercises:</i> • Sit-to-stand exercises 10-20 reps BID • Dancing exercises 30 sec-2 min BID-TID • Stair Climbing 1 working up to 5 flights BID-TID • Light jogging 5-8 minutes BID Medications • NSAIDs—PRN Modalities • Cryotherapy PRN after exercises	Therapeutic Exercises • Leash walks 20-30 min including up to 10-15 min of inclines • Incorporate challenging surfaces to walks, eg snow, sand, tall grass—to increase stifle flexion <i>Note: Ensure radiographic evidence of fracture healing and implant integrity preceding to these exercises:</i> • Sit-to-stand exercises 10-20 reps BID • Dancing exercises 30 sec-2 min BID-TID • Stair Climbing 1 working up to 5 flights BID-TID • Light jogging 5-8 minutes BID Medications • NSAIDs—PRN Modalities • Cryotherapy PRN after exercises	• Goniometry—stifle ROM and other joints if applicable • Re-evaluate thigh muscle girth at 7-8 weeks postoperative • Response to activity and subjective pain level • Lameness score at a stance, walk, and trot • Weight	• Consistent partial weight-bearing on operated limb at a trot • Minimal to no pain with extensive activities

*This protocol is for use by licensed personnel who have been trained in canine rehabilitation. The therapist must keep in mind that these timelines are intended as guidelines and may be influenced by many factors affecting individualized patient care.

**This device is intended to minimize the risk of quadriceps contracture by controlling/preventing stifle extension.

Degenerative Lumbosacral Stenosis—Conservative Management*

Stage	Rehabilitation Clinic Program	Home Program	Outcome Assessment Measures
Mild Pain <u>Goals:</u> No spinal/paraspinal pain Perform at all gait activities without pain Eliminate limb muscle atrophy Strengthen paraspinal muscles	Therapeutic Exercises <i>Introduce gradually as tolerated by the patient</i> - Balance on a 360-degree balance board (all four limbs) or semi-inflated disk (hindlimbs) 5 min SID working up to 10 min BID - Aquatic therapy: Underwater Treadmill (UWTM) walking, 5-10 min SID steady speed working up to 20-30 min BID variable speed - Swimming 5 working up to 20 minutes adding jets 2-3 times per week - Walking working up to jogging through weave poles, gradually decreasing distance between poles, 3 working up to 10 laps through 6-8 poles SID-BID - Cavaletti rails 5 working up to 10 reps, increasing height and decreasing distance, SID-BID - Walking along the length of a foam pad or mattress 5 working up to 10 reps SID-BID - Walking up and down stairs, 1-2 working up to 5-8 flights SID-BID. <i>Add this exercise only if lumbosacral pain is well controlled</i> - Sit-to-stand 5 working up to 15-20 reps SID-BID - Balance with pelvic limbs on a high-traction surface and antebrachii or forepaws balancing on a physioroll or peanut. Apply gradually increasingly challenging displacement forces in multiple directions to encourage walking of the hindlimbs while balancing. 5-10 min SID-BID. <i>Select an appropriately sized physioroll to prevent spinal hyperextension with this activity.</i> <i>Caution: Choose only 2-4 exercises per session. Other exercises may be performed as part of a home exercise program. Doing all exercises may result in over-exertion and pain early in the program</i>	Therapeutic Exercises <i>Introduce gradually as tolerated by the patient</i> - Balance on a moderately unstable surface (all four limbs) or semi-inflated disk (hindlimbs) 5 min SID working up to 10 min BID - Walking working up to jogging in circles and figure eights, gradually decreasing size, 5-10 min SID-BID - Walking in sand, snow, low water or across a foam mattress 5-10 minutes 2-3 times per week. - Sit-to-stand 5 working up to 15-20 reps SID-BID - Walking up and down stairs, 1-2 working up to 5-8 flights SID-BID. <i>Add this exercise only if lumbosacral pain is well controlled</i> - Balance “rides” while standing in a cart, wagon, or wheelbarrow over increasingly variable terrain, 5-10 min SID. <i>Ensure the patient is supported at all times to prevent falling.</i> - Moderate leash walks 20-30 min BID, incorporating 5 minutes of mild-moderate inclines and 2-5 minutes jogging <i>Caution: Choose only 2-4 exercises per session. Doing all exercises may result in over-exertion and pain early in the program.</i>	- Subjective pain level at digital pressure L7-S1, dorsal tail elevation, back hyperextension, and rectal palpation - Hindlimb proprioception and postural reactions - Thigh circumference - Ability to navigate challenging obstacles while jogging/turning

Continued